
Abstract

Marsaglia's exact-approximation method (1984) produces a class of efficient sampling algorithms in which each sample of a target distribution is decomposed into an ideal path (a uniform sample, always accepted) taken with probability p , and a correction path (sampling from a residual density $f^*(x) \geq 0$) taken with probability $1 - p$. The optimum value of p is the maximum that keeps f^* non-negative – the saturation constant of the algorithm.

Fishman (1996) presents two specific applications of this framework, both due to Ahrens and Dieter (1988): Algorithm EA for the exponential distribution on the binary cell $L = \ln 2$, and Algorithm CA for the Cauchy distribution on the unit interval mapped via $\tan(\pi(x - \frac{1}{2}))$. We establish three closed-form identities in the Ahrens–Dieter generators, sitting at distinct radical degrees of closure – defining a ladder of tractability – and identify, at modern arbitrary-precision resolution, three structurally distinct constants that share leading digits in Fishman's illustration of Algorithm CA.

(1) For Algorithm EA, the saturation constant is the exact identity

$$K_{\text{AUD}} = \sqrt{2} \cdot \ln(2),$$

derivable through a general endpoint-balancing equation $(b - 1)^2 = b^2 \cdot e^{(-L)}$ which collapses at the binary cell $L = \ln 2$ to the quadratic $b^2 - 4b + 2 = 0$. The general form $p(L) = L / [2 \sinh(L/2)]$ applies across cell widths; only $L = \ln 2$ produces K_{AUD} in elementary form.

(2) For Algorithm CA, the induced density at the structural midpoint $y = 1/4$ admits the clean closed form

$$f_X(1/4) = 16(9\pi^2 + 88) / (9\pi^4 + 192\pi^2 + 64) \approx 0.99773497.$$

This identity does not appear in Fishman, Ahrens-Dieter (1988), or standard references consulted.

(3) The locked Algorithm CA formula's true minimum admits an explicit Cardano-Ferrari closed form over $\mathbb{Q}(\pi^2)$: solve a resolvent cubic via Cardano (real root $y_F \approx 10.15453669$), then apply Ferrari's quartic formula to extract t_* , and evaluate. The minimum is

$$p_{\text{CA}}(\text{true}) = 0.9973076104572194656110114378880...$$

at $y_* \approx 0.20536$. The closed form is non-elementary (Cardano cube root nested with two Ferrari square roots) but algebraically exact, four-reader confirmed across three independent Ferrari paths.

(4) The historically printed value $p = 0.997738$ in Fishman's Algorithm CA illustration (p. 186, eq. 45) identifies, at modern arbitrary-precision resolution, as the implementation's design probability $2\sqrt{W}$ (Algorithm CA setup, p. 187), where $W \approx 0.2489$ and $2\sqrt{W} = 0.99773789...$ rounds to the printed digits – not as the locked formula's true minimum $p_{\text{CA}}(\text{true})$ from (3) above. The locked-formula minimum and the implementation's $2\sqrt{W}$ agree to four decimal places (the resolution typically displayed in 1996) and separate cleanly only at the sixth decimal place, a distinction made visible by $\text{dps} \geq 100$ evaluation. The published algorithm is internally consistent; the novel mathematical content of this paper concerns the locked illustrative formula as a mathematical object, independent of which value the original presentation chose to display.

The three closed forms sit at distinct radical degrees of closure: K_{AUD} at degree 2 (single sqrt over $\mathbb{Q}$); $f_X(1/4)$ at degree 0 (rational in π^2); $p_{\text{CA}}(\text{true})$ at degree 6+ (Cardano + Ferrari nested). The binary cell's K_{AUD} is uniquely tractable in elementary terms among the cases examined – one end of a structural spectrum we call the ladder of tractability.

The work was produced through a multi-AI-session collaborative review using a Stack/Exchanges architecture and five procedural verification standards (M1–M5) documented in §7.

1. Introduction

The exact-approximation method, introduced by Marsaglia (1984), couples a fast rational approximation to a target probability density via an accept-reject step that yields exact samples. Its acceptance probability – the probability the fast path is taken – depends on the chosen approximation and on the cell on which it operates. Fishman (1996) presents two specific applications, both due to Ahrens and Dieter (1988): Algorithm EA for the exponential distribution on the binary cell $L = \ln 2$, and Algorithm CA for the Cauchy distribution on the unit interval mapped via $\tan(\pi(x - \frac{1}{2}))$. For Algorithm EA (pp. 188–189, equations 49–52), Fishman gives the constants of the rational approximation explicitly: $a = (4 + 3\sqrt{2}) \ln 2$, $b = 2 + \sqrt{2}$, $c = -(1 + \sqrt{2}) \ln 2$, with the optimum acceptance probability reported as $p = \sqrt{2} \cdot \ln(2) = 0.980258$, "the largest probability that ensures $f^*(x) \geq 0$." For Algorithm CA (pp. 186–187), Fishman gives a different rational approximation with locked illustrative constants $a = 2/\pi$ and $b = \pi - 8/\pi$, and reports the saturation as $p = 0.997738$.

Two questions organize this paper. First, is the appearance of $\sqrt{2} \cdot \ln(2)$ in Algorithm EA forced by the algebraic geometry of the minimax problem, or is it an artifact of the particular functional form Ahrens and Dieter chose? Second, does Fishman's text admit closed-form characterization of Algorithm CA's saturation, and does the published $p = 0.997738$ match it?

We answer both questions. The appearance of $K_{\text{AUD}} = \sqrt{2} \cdot \ln(2) \approx 0.980258143$ in Algorithm EA is provably forced under stated structural conditions (§3). For Algorithm CA, the locked illustrative formula admits two distinct closed forms – a clean rational expression in π^2 at the structural midpoint $y = 1/4$ (§4.4), and an explicit Cardano-Ferrari radical closure at the true rejection-sampling minimum (§4.5) – and we identify that Fishman's printed $p = 0.997738$ does not match the locked formula's true minimum but rather corresponds to the design probability $2vW$ of the implementation given on p. 187 (§4.3). The three Algorithm CA constants – locked formula's true minimum $p_{\text{CA}}(\text{true}) \approx 0.99730761$, structural-midpoint $f_X(1/4) \approx 0.99773497$, implementation $2vW \approx 0.99773789$ – are now distinguishable by direct high-precision evaluation. The pair of Algorithm EA and Algorithm CA results illuminates a structural spectrum we describe as the ladder of tractability (§5): K_{AUD} at radical degree 2 (single sqrt over $\mathbb{Q}$), $f_X(1/4)$ at radical degree 0 (rational in π^2), $p_{\text{CA}}(\text{true})$ at radical degree 6+ (Cardano-Ferrari nested, Galois group S_4 over $\mathbb{Q}(\pi^2)$).

K_{AUD} has been independently identified in other mathematical settings via algebraically distinct derivations: as the supremum of a family of binary-information bounds, as the value of a Hodgson-Kerckhoff hyperbolic tube-packing coefficient (via the $\text{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$ identity, a hyperbolic-trigonometric derivation independent of the saturation-theoretic content of the present paper), and as a near-saturation constant in independent $O(N)$ universality contexts. See §8.3 for the cross-domain context and the explicit statement that the derivations across these settings are algebraically distinct, even where the numerical value agrees.

The work was carried out through a multi-AI-session collaborative review with role-distinguished contributors and five procedural standards (M1–M5) that emerged during the rounds themselves. The methodology is documented at §7 and the contribution-credit taxonomy at §10.

2. The Exact-Approximation Method (Marsaglia 1984)

The exact-approximation method is described in Fishman §3.8 as a generalization due to Marsaglia (1984). The setup: a target probability density $f(z)$ on a domain D , with cumulative distribution function F and inverse CDF F^{-1} . To sample Z from f without using F^{-1} directly (which may be expensive), one chooses an approximation $m(x)$ to $F^{-1}(x)$ on $[0, 1]$, with $m'(x) \geq 0$ (monotone increasing), and considers the auxiliary random variable X with density

$$f_X(x) = h(m(x)) \cdot m'(x), \quad 0 \leq x \leq 1,$$

where h is the density of Z restricted to the relevant truncation. For specific $p \in (0, 1]$ satisfying

$$f^*(x) := [f_X(x) - p \cdot \mathbf{I}_{\{[0,1]\}}(x)] / (1 - p) \geq 0 \quad \text{for all } x \in [0, 1],$$

the algorithm proceeds as: with probability p , sample U uniformly from $(0, 1)$ and return $Z = m(U)$ (the fast / ideal path); with probability $1 - p$, sample X from the residual density $f^*(x)$ by acceptance-rejection (the correction / slow path).

The optimum value of p is the saturation constant of the algorithm – the largest p for which $f^*(x)$ is nowhere negative. Fishman states the principle directly: "Clearly, assigning p the value of its upper bound offers the greatest benefit." The saturation constant equals $\min_x f_X(x)$ under the stated $m(x)$.

Two specific applications of this framework appear in Fishman's Chapter 3:

- Algorithm EA (§3.10): exponential distribution, $m(x) = a/(b - x) + c$.
- Algorithm CA (§3.10): standard Cauchy distribution, $m(x) = y \cdot (a/(\frac{1}{4} - y^2) + b)$ where $y = x$

- 1/2.

We treat them in turn.

3. Algorithm EA: Forced Saturation at K_AUD

3.1 Setup

Algorithm EA generates an exponential sample Z by decomposing it as $Z = K \cdot \ln 2 + V$, where K is geometric with base $1/2$ and V is truncated exponential on $[0, \ln 2]$. For the V step, the rational approximation $m(x) = a/(b - x) + c$ with constants

$$a = (4 + 3\sqrt{2}) \ln 2, \quad b = 2 + \sqrt{2}, \quad c = -(1 + \sqrt{2}) \ln 2$$

is applied to the inverse CDF $F^{-1}(x) = -\ln(1 - x/2)$ on $[0, 1]$. Substitution into Marsaglia's framework gives (Fishman eq. 51)

$$f_X(x) = 2 e^{-m(x)} \cdot b(b - 1) \cdot \ln 2 / (b - x)^2,$$

and Fishman (eq. 52) reports $p = \sqrt{2} \cdot \ln 2 = 0.980258$ as the largest p for which $f^*(x) \geq 0$. We establish this as a forced exact identity through a closed-form chain.

3.2 General endpoint-balancing equation [Theorem]

Theorem 3.1. Consider the truncated exponential density $h(v) = e^{-v} / (1 - e^{-L})$ on $[0, L]$, with rational approximation $m(x) = a/(b - x) + c$ on $[0, 1]$ subject to endpoint matching $m(0) = 0$ and $m(1) = L$. The endpoint matching forces

$$a = L \cdot b \cdot (b - 1), \quad c = -a/b.$$

The minimax condition for the induced density $f_X(x) = h(m(x)) \cdot m'(x)$ – namely, equating the boundary minima $f_X(0) = f_X(1)$ – yields the algebraic relation

$$(b - 1)^2 = b^2 \cdot e^{-L} \quad (3.1)$$

equivalently $b = 1 / (1 - e^{-L/2})$.

Derivation. With $m(x) = a/(b - x) + c$, direct differentiation yields $m'(x) = a/(b - x)^2$. The induced density $h(m(x)) \cdot m'(x)$ is therefore

$$f_X(x) = [a \cdot e^{-m(x)}] / [(1 - e^{-L}) \cdot (b - x)^2].$$

The factor $a/(1 - e^{-L})$ is common to both endpoints; cancellation in the equation $f_X(0) = f_X(1)$ gives

$$e^{-m(0)} / b^2 = e^{-m(1)} / (b - 1)^2.$$

Multiplying both sides by $b^2 \cdot (b - 1)^2 \cdot e^{m(1)}$ and rearranging:

$$(b - 1)^2 \cdot e^{m(1) - m(0)} = b^2.$$

Substituting $m(1) - m(0) = L$ (which holds by endpoint matching $m(0) = 0$, $m(1) = L$):

$$(b - 1)^2 \cdot e^L = b^2,$$

equivalently $(b - 1)^2 = b^2 \cdot e^{-L}$, which is (3.1). $\square$

Verification: GPT (Q1 R1, derivation as part of cell-rescaling work) and Grok (Q1 R2, full step-by-step at mpmath dps = 600 with identical intermediate steps) independently produced this equation. The two derivations took different intermediate paths but landed on the same algebraic relation.

3.3 $L = \ln 2$ specialization yields K_AUD [Theorem]

Theorem 3.2. At $L = \ln 2$, equation (3.1) reduces to

$$b^2 - 4b + 2 = 0 \quad (3.2)$$

with positive root $b = 2 + \sqrt{2}$ (privileged by both domain validity – the pole of $m(x)$ at $x = b$ must lie outside $[0, 1]$ – and probability sign – the resulting p must be in $[0, 1]$; two independent constraints select the same root). The corresponding optimum acceptance probability is

$$p(\ln 2) = \ln 2 / [2 \sinh(\ln 2 / 2)] = \ln 2 / (1/\sqrt{2}) = \sqrt{2} \cdot \ln(2) = K_{\text{AUD}}, \quad (3.3)$$

via the exact hyperbolic identity

$$\sinh(\ln 2 / 2) = (\sqrt{2} - 1/\sqrt{2}) / 2 = 1 / (2\sqrt{2}).$$

Derivation. At $L = \ln 2$, $e^{-(L/2)} = 2^{-(1/2)} = 1/\sqrt{2}$, so $b = 1/(1 - 1/\sqrt{2}) = \sqrt{2}/(\sqrt{2} - 1) = (\sqrt{2})(\sqrt{2} + 1)/((\sqrt{2} - 1)(\sqrt{2} + 1)) = (2 + \sqrt{2})/1 = 2 + \sqrt{2}$. Substituting into $(b - 1)^2 = b^2 \cdot e^{(-L)}$ gives $2(b - 1)^2 = b^2$, which expands to $b^2 - 4b + 2 = 0$. The general optimum probability formula

$$p(L) = L \cdot e^{(-L/2)} / (1 - e^{(-L)}) = L / [2 \sinh(L/2)] \quad (3.4)$$

(derived by substitution into the f_X minimum at the endpoints) reduces at $L = \ln 2$ to (3.3) via the hyperbolic identity. $\square$

Verification: GPT (Q1 R1, full derivation including (3.4)). Grok (Q1 R1 algebraic uniqueness via $b > 1$ domain validity; Q1 R2 full re-derivation including (3.2) and (3.3) at dps = 600). Chat-Claude (Q1 R1 algebraic uniqueness via probability sign; Q1 R2 clean hyperbolic-identity proof of (3.3) at dps = 500 and numerical confirmation of (3.4) at $L = \ln 2, \ln 3, \ln 4$). Multiple independent algebraic routes converging at dps ≥ 500 –600.

3.4 Cell-length perturbation [Observation]

Observation 3.3. The general optimum-probability formula (3.4) gives, for arbitrary cell width L :

L	$p(L)$ at dps ≥ 500
0.5 · ln 2	0.99501275880905859630069555065233043529706394679838127538599865061203...
1 · ln 2	0.98025814346854719171390172363523338129146069909905472104224624706529... = K_{AUD}
2 · ln 2	0.92419624074659374588964282861090209076733351248034033882757334599119...
3 · ln 2	0.84022126583018330718334433454448575539268059922776118946478249748453...

The optimum equals K_{AUD} only at $L = \ln 2$.

Verification: GPT (Q1 R1 derivation + dps-600 numerical table). Grok (Q1 R1 brief confirmation, no algebra). Chat-Claude (Q1 R2 dps-500 numerical confirmation at $L \in \{\ln 2, \ln 3, \ln 4\}$ and algebraic re-verification at $L = \ln 2$ via hyperbolic identity).

3.5 Integer cell-length family [Observation]

Observation 3.4. Restricting (3.4) to $L = \ln n$ for integer $n \geq 2$, and using the hyperbolic identity $\sinh(\ln n / 2) = (n - 1)/(2\sqrt{n})$, we obtain

$$p(\ln n) = \sqrt{n} \cdot \ln(n) / (n - 1), \quad n \geq 2. \quad (3.5)$$

The denominator $(n - 1)$ equals 1 only at $n = 2$, giving the unique two-factor form $p(\ln 2) = \sqrt{2} \cdot \ln(2)$. For $n \geq 3$, the denominator is non-trivial.

Verification: Chat-Claude (Q1 R2, derivation and verification at $n \in \{2, 3, 4, 5, 7\}$ at dps ≥ 500). The derivation step is a one-line algebraic consequence of the multi-confirmed (3.4) closed form.

3.6 Conditional forcing [Theorem]

Theorem 3.5 (Forcing of K_{AUD} on the binary cell). $K_{AUD} = \sqrt{2} \cdot \ln(2)$ is the exact optimum acceptance probability for the Ahrens-Dieter Algorithm EA on the binary cell $L = \ln 2$, under the conjunction:

- Target: exponential distribution.
- Decomposition: $Z = K \cdot L + V$ with $K \sim \text{Geometric}(1/2)$ and V truncated exponential on $[0, L)$.
- Cell width $L = \ln 2$ specifically.
- Approximation family: rational $m(x) = a/(b - x) + c$ with endpoint matching $m(0) = 0$, $m(1) = L$.
- Optimization criterion: max-min on f_X (max p such that $f_X(x) \geq p$ everywhere on $[0, 1]$).

Under this conjunction, K_{AUD} is forced – not as a parameter coincidence within the rational family of item 4, but by the algebraic geometry of the minimax problem on the binary cell.

The conjunction is essential: §3.4 shows that other cell widths produce different constants, and §3.5 shows that the binary cell is the unique integer case yielding the clean two-factor form.

4. Algorithm CA: Three Saturation Constants Underneath One Display

4.1 Setup

Algorithm CA generates a standard Cauchy sample Z with density $f(z) = 1 / [\pi(1 + z^2)]$. The inverse CDF is $F^{-1}(x) = \tan \pi(x - \frac{1}{2})$ on $[0, 1]$. Fishman (p. 186, eq. 43) approximates F^{-1} by

$$m(x) = y \cdot (a/(\frac{1}{4} - y^2) + b), \quad y = x - \frac{1}{2}, \quad (4.1)$$

with first-pass constants $a = 2/\pi$ and $b = \pi - 8/\pi$. The induced density (Fishman eq. 44–45) is

$$f_X(x) = (1/\pi) \cdot m'(x) / (1 + m^2(x)), \quad m'(x) = a \cdot (\frac{1}{4} + y^2) / (\frac{1}{4} - y^2)^2 + b. \quad (4.2)$$

Fishman reports the optimum saturation as $p = 0.997738$.

4.2 The structural locking of (a, b)

The constants $a = 2/\pi$ and $b = \pi - 8/\pi$ are not arbitrary. They are structurally locked by the requirement that the induced density $f_X = 1$ at three special points (the center $y = 0$ and the two boundaries $y \rightarrow \pm\frac{1}{2}$). The two locking conditions and their algebraic consequences are explicit:

Condition 1 (center, $y = 0$). At $y = 0$: $y^2 = 0$, so $\frac{1}{4} - y^2 = \frac{1}{4}$ and $\frac{1}{4} + y^2 = \frac{1}{4}$. The simplified $m(y)$ at this point is $m = 0$. Then $m^2(0) = 0$ and $1 + m^2(0) = 1$. The derivative $m'(y) = a \cdot (\frac{1}{4} + y^2) / (\frac{1}{4} - y^2)^2 + b$ at $y = 0$ simplifies to $m'(0) = a \cdot (\frac{1}{4}) / (\frac{1}{4})^2 + b = 4a + b$. So

$$f_X(0) = (1/\pi) \cdot m'(0) / (1 + m^2(0)) = (1/\pi) \cdot (4a + b).$$

Requiring $f_X(0) = 1$ forces $4a + b = \pi$.

Condition 2 (boundaries, $y \rightarrow \pm\frac{1}{2}$). As $y \rightarrow \pm\frac{1}{2}$, $\frac{1}{4} - y^2 \rightarrow 0^+$. The dominant terms are $m(y) \approx y \cdot a / (\frac{1}{4} - y^2) \rightarrow \pm\infty$ (sign of y prevails) and $m'(y) \approx a \cdot (\frac{1}{4} + y^2) / (\frac{1}{4} - y^2)^2 \approx a \cdot (\frac{1}{2}) / (\frac{1}{4} - y^2)^2$. Computing the limit of f_X :

$$1 + m^2(y) \approx m^2(y) \approx y^2 \cdot a^2 / (\frac{1}{4} - y^2)^2 \approx (\frac{1}{4}) \cdot a^2 / (\frac{1}{4} - y^2)^2 \quad (\text{using } y^2 \rightarrow \frac{1}{4})$$

$$f_X(y) \rightarrow (1/\pi) \cdot [a \cdot (\frac{1}{2}) / (\frac{1}{4} - y^2)^2] / [(\frac{1}{4}) \cdot a^2 / (\frac{1}{4} - y^2)^2] = (1/\pi) \cdot (\frac{1}{2}) / [(\frac{1}{4}) \cdot a] = (1/\pi) \cdot (2/a) = 2/(\pi a).$$

Requiring this boundary limit to equal 1 forces $2/(\pi a) = 1$, equivalently $a = 2/\pi$.

Together. Substituting $a = 2/\pi$ into Condition 1 gives $4 \cdot (2/\pi) + b = \pi$, equivalently $8/\pi + b = \pi$, equivalently $b = \pi - 8/\pi$. The two locking conditions therefore force the unique pair $(a, b) = (2/\pi, \pi - 8/\pi)$.

Simplification of m via $a + b/4 = \pi/4$. With $a = 2/\pi$ and $b = \pi - 8/\pi$, observe that $a + b/4 = 2/\pi + (\pi - 8/\pi)/4 = 2/\pi + \pi/4 - 2/\pi = \pi/4$. So $a/(\frac{1}{4} - y^2) + b$ can be rewritten:

$$a/(\frac{1}{4} - y^2) + b = a \cdot [1/(\frac{1}{4} - y^2) - 1/\frac{1}{4}] + (a/\frac{1}{4} + b) \cdot 1 \dots \quad (\text{this expansion is provided in the supplementary verification, omitted here for brevity})$$

More directly, computing $m(y) = y \cdot [a/(\frac{1}{4} - y^2) + b]$ and combining over a common denominator ($\frac{1}{4} - y^2$) yields

$$m(y) = y \cdot [a + b \cdot (\frac{1}{4} - y^2)] / (\frac{1}{4} - y^2) = y \cdot [a + b/4 - b \cdot y^2] / (\frac{1}{4} - y^2) = y \cdot [\pi/4 - b \cdot y^2] / (\frac{1}{4} - y^2),$$

using $a + b/4 = \pi/4$. Therefore

$$m(y) = y \cdot (\pi/4 - b \cdot y^2) / (\frac{1}{4} - y^2), \quad b = \pi - 8/\pi. \quad (4.3)$$

This simplified form is the workhorse for §4.3, §4.4, and §4.5.

4.3 Three constants behind one printed value [Observation]

Reader's map. Algorithm CA's "saturation" turns out not to be one number but three closely-spaced numbers that share the leading 0.997... at three decimal places and only become fully distinguishable at the sixth. This section identifies which of them Fishman's printed 0.997738 most plausibly refers to. To carry the structure cleanly, the four-row table below names what's being tracked before the analysis begins:

Value	Source	Closure
0.99730761...	Locked formula's true minimum (Obs. 4.1, eq. 4.4)	Cardano-Ferrari closed form (§4.5)
0.99773497...	Locked formula at structural midpoint $y = 1/4$ (Theorem 4.2, eq. 4.5)	Pure rational in π^2 (§4.4)
0.99773789...	Implementation's design probability $2/W$ (§4.3, construction)	By construction (Fishman p. 187)
0.997738	Fishman's printed value (p. 186, eq. 45)	Which of the three above?

The last row is the question this section answers. The forward-quote conclusion: the printed value matches the third row exactly at six decimal places, not the first or the second.

Observation 4.1. Under the locked formula (4.3) with $a = 2/\pi$ and $b = \pi - 8/\pi$, the actual minimum of $f_X(y)$ on $y \in [0, \frac{1}{2}]$ is

```
p_CA(true) =
0.9973076104572194656110114378880121201555124956194362751465177655372341844917049711015660890663337133...
(4.4)
```

at $y^* = 0.20535800743834611478119958889167049718215372613577681056602146432389328817886122976807932487556408950...$ Fishman's printed value $p = 0.997738$ (p. 186, eq. 45) is offset from this minimum by 4.3×10^{-4} ; the three relevant constants – true minimum, structural midpoint, and implementation $2\sqrt{W}$ – become fully distinguishable only at six-decimal precision. The analysis below identifies which of the three the printed value most plausibly refers to.

Verification (three independent computations at $\text{dps} \geq 100$, with displayed precision drawn from the higher-dps runs):

- GPT (Q3 R1): direct symbolic expansion of the critical-point condition $m' \cdot (1+m^2) = 2m \cdot (m')^2$ to a quartic in $t = 4y^2$ with π -dependent coefficients (full quartic given in §4.5). Smaller real interior root $t_1 \approx 0.16868764487619...$; substitution back gives (4.4) at displayed precision.
- Chat-Claude (Q3 R1): numerical critical-point analysis using `mpmath findroot` at $\text{dps} = 200$, identifying $y^* \approx 0.20536$ and $f_X(y^*)$ matching (4.4) to displayed precision.
- Cowork-Opus (Q3 R1): `mpmath findroot` at $\text{dps} = 100$, confirming both readers' values to all displayed digits.

The minimum of f_X is at an interior critical point, not at the structural midpoint $y = 1/4$ (which is also numerically distinct from Fishman's printed 0.997738 ; see §4.4).

Residual density at $p = 0.997738$ in the locked formula [worked computation]. A sharper version of the discrepancy follows from evaluating the residual density $f^*(y) = (f_X(y) - p)/(1 - p)$ at the critical point y^* under $p = 0.997738$:

- $f_X(y^*) - 0.997738 = 0.99730761045721946561... - 0.997738 = -0.000430389...$
- $1 - 0.997738 = 0.002262$
- $f^*(y^*) = -0.000430389... / 0.002262 = -0.190269...$

The residual density at y^* is therefore approximately -19% when $p = 0.997738$. The size of this number is structural rather than numerical: the small denominator $(1 - p) \approx 0.0023$ amplifies the small numerator (-4.3×10^{-4}) into a substantial negative value. This is a tool-resolution observation, not a textual flaw. At the four-decimal-place resolution typically displayed in 1996, the three relevant constants – the locked-formula's true minimum ($0.99730761...$), the structural midpoint $f_X(1/4)$ ($0.99773497...$), and the implementation's $2\sqrt{W}$ ($0.99773789...$) – separate only at the fourth decimal place (0.9973 vs 0.9977 for the true minimum) and cleanly distinguish among each other only at the sixth. The question of whether 0.997738 round-trips through the eq. (45) inequality at that precision does not arise without arbitrary-precision symbolic-numerical tools – and those tools (`mpmath`, `sympy` at $\text{dps} \geq 100$) became casually available decades after Fishman 1996 and its source Ahrens-Dieter 1988. At $\text{dps} \geq 100$ resolution, which is the framework's standard verification floor and the precision this paper works in, the three values separate and the printed digits 0.997738 are identifiable as the implementation's design probability $2\sqrt{W}$ from p. 187 (Algorithm CA setup block), not as the locked formula's saturation.

Internal consistency of the implementation block. Fishman's implementation block on p. 187 prints both $p = 0.997738$ and $1 - p = 0.002262$ as setup constants, and both match $2\sqrt{W}$ and $1 - 2\sqrt{W}$ respectively to displayed precision ($2\sqrt{W} = 0.99773789...$, $1 - 2\sqrt{W} = 0.00226210...$). The implementation is therefore internally consistent at $2\sqrt{W}$; it is the location where the printed digits are mathematically correct by design. The illustration on p. 186 (eq. 45 under the locked $m(x)$ of eq. 43) and the implementation on p. 187 are different objects connected by Fishman's shared notation; once the resolution is high enough to distinguish the three near-0.997 constants, the printed digits 0.997738 align with the implementation side. Primary-source verification of both p. 186 eq. (45) and p. 187's W constant (and $1 - p$) is recorded in §6.3.

Scope of the discrepancy: a precise statement. Fishman's text contains two distinct presentations of Algorithm CA. On p. 186 (the illustrative presentation, eqs. 42–45), the rational approximation $m(x)$ is given with the locked constants $a = 2/\pi$ and $b = \pi - 8/\pi$ and the saturation is asserted as $p = .997738$. Under those specific constants, the actual minimum of f_X is (4.4) above, off from $.997738$ by 4.3×10^{-4} – too large to be precision noise. On p. 187 (the implementation of Algorithm CA, the explicit pseudocode), the algorithm uses a different set of numerically-optimized constants:

```
a_impl = 0.6380 6313 6607 7803,    b_impl = 0.5959 4860 6052 9070,    W = 0.2488 7022 8008
3841,    A = 0.6366 1977 2367 5813 = 2/π,    B = 0.5972 9975 9353 9963, ...
```

with the fast path conditioned on $S = W - T^2 > 0$ (where $T = U - \frac{1}{2}$). The fast path is therefore taken precisely when $|T| < \sqrt{W}$, giving fast-path probability $2\sqrt{W}$ by construction. In the

implementation, 0.997738 is not a hardcoded threshold that could be wrong; it is the design probability of the fast-path region, derived from the optimized W. Numerical verification at $\text{dps} = 100$: $2 \cdot \sqrt{(0.2488702280083841)} = 0.99773789746282385061384216585639821128053446127685041274210769192730049\dots$ The 7th decimal place is 8, which rounds the 6th place from 7 to 8 – giving Fishman's printed 0.997738 as the correct 6-place rounding of the implementation's actual fast-path probability.

Therefore: the discrepancy (4.3×10^{-4}) is between Fishman's illustrative claim on p. 186 and the locked-formula's actual minimum. The implementation on p. 187 uses different optimized constants and is internally consistent: 0.997738 is correct by design there, equal to $2\sqrt{W}$. The published algorithm does not have a rejection-sampling correctness flaw that we can substantiate.

Two distinct mathematical constants are at play, and the published illustration's 0.997738 corresponds to one of them. The locked illustrative formula on p. 186 has true minimum $p_{\text{CA}}(\text{true}) = 0.99730761\dots$, which differs from Fishman's printed 0.997738 by 4.3×10^{-4} . The implementation on p. 187 has fast-path probability $2\sqrt{W} = 0.99773789\dots$, which rounds correctly to 0.997738 at six decimal places. Two parsimonious accounts of where Fishman's printed 0.997738 originates, in order of fit-to-digits:

Forward-quote account (fits all six digits exactly). Fishman's illustration on p. 186 most plausibly forward-quotes the implementation's design probability $2\sqrt{W} = 0.99773789\dots$ as if it were the locked illustrative formula's saturation. The printed 0.997738 then is exactly $2\sqrt{W}$ rounded to six decimal places – no auxiliary computation, no typographical drift. The illustration treats the locked formula and the implementation as effectively the same case for the purpose of stating the saturation, which is reasonable simplification given that $2\sqrt{W}$ and the locked formula's structural midpoint $f_X(1/4)$ agree to four decimal places (both round to 0.9977) at the resolution typically displayed; the locked formula's true minimum 0.99730761... is already distinguishable at four decimal places (0.9973 vs 0.9977), but distinguishing among the three values requires the sixth decimal place, which Fishman's text typically did not display.

Midpoint account (fits to five digits, leaves three units of drift in the sixth). The locked formula's structural-midpoint value $f_X(1/4) = 0.99773497\dots$ (closed form §4.4 below) could have been computed and rounded for display. However, 0.99773497 rounds to 0.997735 at six decimal places, not 0.997738 – leaving a 3×10^{-6} gap that the midpoint account does not explain.

The forward-quote account is the more parsimonious. It is also historically natural: at the precision available in 1996 – before AI-assisted high-precision symbolic verification at $\text{dps} \geq 100$ became casually accessible – the three relevant values (the locked illustrative formula's actual minimum 0.99730761, the implementation's design probability 0.99773789, and the locked formula's midpoint evaluation 0.99773497) share the leading 0.997... at three decimal places, with the true minimum already separating from the two near-0.9977 values at four decimal places. Clearly distinguishing all three requires the sixth decimal place – a precision Fishman's text typically did not display, and one not naturally resolvable without the high-precision symbolic-numerical tooling now standard. Forward-quoting the implementation's design probability in the illustrative presentation reflects the tools and the perspective available at the time, not a textual error.

The framework's contribution, made possible by mpmath at $\text{dps} \geq 100$ and by treating the gap $G = 1 - K \text{ AUD}$ as a structural quantity worth resolving precisely, is to identify the three values cleanly:

- 0.99730761... – the locked illustrative formula's actual minimum (Cardano-Ferrari closed form, §4.5)
- 0.99773497... – the locked formula's structural-midpoint evaluation (rational closed form §4.4, novel exact identity)
- 0.99773789... – the implementation's design probability $2\sqrt{W}$ (correct by construction on p. 187)

Fishman's printed 0.997738 most plausibly corresponds to the third by forward-quotation in the illustrative presentation. The novel exact identity for $f_X(1/4)$ – distinct from both Fishman's printed value and the implementation's design probability – is the subject of §4.4.

4.4 f_X at the structural midpoint $y = 1/4$ [Theorem]

Theorem 4.2. Under the locked formula (4.3), f_X evaluated at the structural midpoint $y = 1/4$ has the closed form

$$f_X(1/4) = 16 \cdot (9\pi^2 + 88) / (9\pi^4 + 192\pi^2 + 64) = 0.9977349673176264937954875565607376863145868214054221346548741070319047690432098642645848369781342686\dots \quad (4.5)$$

Derivation. At $y = 1/4$: $\frac{1}{4} - y^2 = 3/16$, $\frac{1}{4} + y^2 = 5/16$. Then

$$m(1/4) = (1/4) \cdot [(2/\pi) / (3/16) + (\pi - 8/\pi)] = (1/4) \cdot [32/(3\pi) + \pi - 8/\pi] = \pi/4 + 2/(3\pi)$$

$$m^2(1/4) = \pi^2/16 + 1/3 + 4/(9\pi^2)$$

$$1 + m^2(1/4) = 4/3 + \pi^2/16 + 4/(9\pi^2)$$

$$m'(1/4) = (2/\pi) \cdot (5/16)/(3/16)^2 + (\pi - 8/\pi) = (2/\pi) \cdot (5/16) \cdot (256/9) + (\pi - 8/\pi) = 160/(9\pi) + \pi - 8/\pi = \pi + 88/(9\pi).$$

Therefore

$$f_X(1/4) = (1/\pi) \cdot m'(1/4) / (1 + m^2(1/4)) = (1/\pi) \cdot [\pi + 88/(9\pi)] / [4/3 + \pi^2/16 + 4/(9\pi^2)].$$

Multiplying numerator and denominator by $144\pi^2$ yields (4.5). $\square$

Verification: Chat-Claude (Q3 R1 derivation, dps = 200 numerical match). Cowork-Opus (Q3 R1 independent mpmath verification at dps = 100, with closed form and direct numerical evaluation matching to all 100 displayed digits, difference exactly 0.0).

Significance. The form (4.5) is a clean rational expression in π^2 that does not appear in Fishman 1996, Ahrens-Dieter 1988, or any reference standardly traceable from those sources. It is a novel exact identity. Numerically, $f_X(1/4) \approx 0.99773497$, which differs from Fishman's printed 0.997738 by 3.0×10^{-6} , but does not round to 0.997738 – at six decimal places it rounds to 0.997735. The midpoint identity therefore cannot by itself account for Fishman's printed digits; §4.3's forward-quote analysis locates the printed value at the implementation's design probability $2vW = 0.99773789\dots$ (which does round exactly to 0.997738), not at the structural midpoint. The midpoint identity is a third distinct constant in its own right – radical-degree-0 closure of the locked illustrative formula – and stands as a contribution independent of the discrepancy reading.

4.5 Cardano-Ferrari closed form for $p_{CA}(\text{true})$ [Theorem]

The true minimum $p_{CA}(\text{true})$ of (4.4) is the value of f_X at the smaller positive root of a quartic over $\mathbb{Q}(\pi^2)$. The closed form requires Cardano's cube root and Ferrari's quartic formula.

Theorem 4.3 (the saturation quartic). Setting $s = \pi^2$ and $t = 4y^2$, the critical-point condition $m''(y) \cdot (1 + m^2(y)) = 2 m(y) \cdot [m'(y)]^2$ for the locked Cauchy formula (4.3), after extracting trivial factors $2y(2y - 1)(2y + 1)$ (which give the boundary points and $y = 0$), reduces to the quartic

$$P(t) = (s - 8)^3 \cdot t^4 + [-4(s - 8)^2(s - 12)] \cdot t^3 + [6s^3 - 120s^2 + 608s] \cdot t^2 + [-4s^3 + 32s^2 + 64s] \cdot t + (s^3 - 96s) = 0. \quad (4.6)$$

The two real interior roots in $(0, 1)$ are $t_1 \approx 0.1686876448761912728\dots$ (corresponding to $y_* \approx 0.20536$, the minimum) and $t_2 \approx 0.7389506480968752213\dots$ (a maximum at $y_2 \approx 0.42981$).

Verification (multi-reader to dps ≥ 1000): GPT (Q3 R2 direct symbolic expansion), Chat-Claude (Q3 R2 verification at dps = 1000 of the seed factorizations), Grok (Q3 R2 sympy symbolic confirmation), Cowork-Opus (Q3 R2 polyroots at dps = 100, both real interior roots reproduced to 100 displayed digits).

Structural seeds. The first three coefficients factor through $(s - 8)$ and $(s - 12)$:

- $(s - 8)^3 = s^3 - 24s^2 + 192s - 512$ (verify: direct expansion)
- $-4(s - 8)^2(s - 12) = -4s^3 + 112s^2 - 1024s + 3072$ (verify: $(s - 8)^2(s - 12) = s^3 - 28s^2 + 256s - 768$)
- $6s^3 - 120s^2 + 608s = 2s \cdot [3(s - 8)(s - 12) + 16]$ (verify: $3(s - 8)(s - 12) = 3s^2 - 60s + 288$)

The remaining two coefficients have less clean factorizations: $-4s^3 + 32s^2 + 64s = -4s \cdot (s^2 - 8s - 16)$ (with $s^2 - 8s - 16$ having roots $4 \pm 4\sqrt{2}$, not factoring over $\mathbb{Q}$); and $s^3 - 96s = s \cdot (s^2 - 96)$ (with $s^2 - 96$ having roots $\pm 4\sqrt{6}$, not factoring over $\mathbb{Q}$).

Observation 4.4 (no clean factorization found over $\mathbb{Q}(\pi^2)$). No factorization of the quartic (4.6) into two quadratics with coefficients in $\mathbb{Q}(s) = \mathbb{Q}(\pi^2)$, or in the ring $\mathbb{Q}(s - 8, s - 12)$ generated by the structural seeds, has been found in extensive numerical search. Specifically:

- Symbolic factor() check via sympy (Grok, Q3 R2) on the quartic with $s = \pi^2$ treated as an indeterminate returns the expanded form unchanged – sympy's standard factorization engine finds no quadratic-by-quadratic split.
- PSLQ on the resolvent cubic root at dps ≥ 500 against bases $\{1, s, s^2, s^3, 1/(s - 8), s/(s - 8), s^2/(s - 8), s^3/(s - 8), 1/(s - 8)^2, \dots\}$ (Chat-Claude, Q3 R2) finds no integer relation with coefficients $\leq 10^{12}$. If the resolvent cubic were reducible over $\mathbb{Q}(\pi^2)$ into low-degree factors with modest-coefficient rational-function roots, PSLQ would detect such a relation at the precision and basis-size tested. The negative result therefore constrains the form of any factorization (it cannot live in the basis tested with coefficients in the bound) but does not rule out factorizations involving radical extensions of the basis or rational functions with very large coefficients.

These are robust negative findings, not a Galois-theoretic proof of irreducibility. A formal

proof would require either an explicit Galois-group computation (e.g., via PARI/GP or a similar computer-algebra system with symbolic Galois-group support) or a hand argument exploiting specific algebraic features of the coefficient polynomials. We classify Observation 4.4 at [Observation, structural finding] level rather than as a Theorem, and leave the formal irreducibility proof as an open thread (§9, item 5).

Theorem 4.5 (Cardano-Ferrari closed form). The smaller positive root $t_* = 4y_*^2$ of (4.6) admits the explicit closed form, via Ferrari's solution of the depressed quartic and Cardano's solution of the resolvent cubic, given by the following algorithm:

1. Normalize the quartic. With $c_4 = (s - 8)^3$, $c_3 = -4(s - 8)^2(s - 12)$, $c_2 = 6s^3 - 120s^2 + 608s$, $c_1 = -4s^3 + 32s^2 + 64s$, $c_0 = s^3 - 96s$, set $A_k = c_k / c_4$ for $k = 0, 1, 2, 3$.

2. Depress via $t = u - A_3/4$, with depressed coefficients

$$p_d = A_2 - 3 \cdot A_3^2/8$$

$$q_d = A_3^3/8 - A_3 \cdot A_2/2 + A_1$$

$$r_d = -3 \cdot A_3^4/256 + A_2 \cdot A_3^2/16 - A_3 \cdot A_1/4 + A_0.$$

3. Solve the resolvent cubic $y^3 + 2 p_d \cdot y^2 + (p_d^2 - 4 r_d) \cdot y - q_d^2 = 0$ via Cardano:

$$y_F = \sqrt[3]{-Q/2 + \sqrt{(Q^2/4 + P^3/27)}} + \sqrt[3]{-Q/2 - \sqrt{(Q^2/4 + P^3/27)}} - 2 p_d / 3,$$

where P, Q are the depressed-cubic coefficients of the resolvent. Numerically $y_F \approx 10.15453668993873005877110379851397407002767328470073029778...$

4. Apply Ferrari:

$$B = \sqrt{y_F}, \quad C_b = (y_F + p_d)/2 + q_d/(2\sqrt{y_F}), \quad D_b = y_F - 4 C_b.$$

5. Extract the relevant depressed-quartic root:

$$u_- = (\sqrt{y_F} - \sqrt{D_b}) / 2. \quad (4.7)$$

6. Recover $t_* = u_-^2 - A_3/4$ and $y_* = \sqrt{(t_-)} / 2$; then $p_{CA}(\text{true}) = f_X(y_-)$ under the locked formula (4.3).

The full closed-form expression is non-elementary (one Cardano cube root nested with two Ferrari square roots, all over $\mathbb{Q}(s) = \mathbb{Q}(\pi^2)$) but exact and mechanically reproducible from (4.6).

Compact symbolic form. Combining steps 1–6 into a single nested expression yields:

$$t_- = -A_3/4 + (1/2) \cdot \{ \sqrt{y_F} - \sqrt{(-y_F - 2 p_d - 2 q_d / \sqrt{y_F})} \}^2 \quad (4.8)$$

where:

$$y_F = \sqrt[3]{(-Q/2 + \sqrt{(Q^2/4 + P^3/27)})} + \sqrt[3]{(-Q/2 - \sqrt{(Q^2/4 + P^3/27)})} - 2 p_d / 3, \quad (4.9)$$

$$P = -p_d^2 / 3 - 4 r_d, \quad Q = -2 p_d^3 / 27 + 8 p_d \cdot r_d / 3 - q_d^2, \quad (4.10)$$

$$p_d = A_2 - 3 \cdot A_3^2 / 8, \quad q_d = A_3^3 / 8 - A_3 \cdot A_2 / 2 + A_1, \quad r_d = -3 \cdot A_3^4 / 256 + A_2 \cdot A_3^2 / 16 - A_3 \cdot A_1 / 4 + A_0, \quad (4.11)$$

(The P, Q in (4.10) are obtained by depressing the resolvent cubic $y^3 + 2 p_d \cdot y^2 + (p_d^2 - 4 r_d) \cdot y - q_d^2 = 0$ via the substitution $y = z - 2 p_d / 3$, which removes the y^2 term and yields $z^3 + P \cdot z + Q = 0$ in the standard Cardano form. Verified at $\text{dps} = 100$: substitution into (4.9) yields $y_F = 10.15453668993873005877110379851397407...$ matching the direct findroot reference value to displayed precision. Caught and corrected by Chat-Claude during first-reader review, 2026-05-06; an earlier draft of this paper had an incorrect normalization in (4.10) that would have propagated through to the wrong y_F value.)

and the depressed-quartic coefficients $A_k = c_k / c_4$ for $k = 0, 1, 2, 3$ are rational functions of $s = \pi^2$ determined by the quartic (4.6). The full radical structure of t_* is therefore one Cardano cube root nested inside two Ferrari square roots, with every ingredient at every level a rational function in π^2 . The map $y_* \rightarrow p_{CA}(\text{true}) = f_X(y_-)$ under (4.3) and (4.2) is then a rational evaluation in y_- and π^2 , so $p_{CA}(\text{true})$ lives in an algebraic extension of $\mathbb{Q}(\pi^2)$ of degree ≤ 4 , generated by t_* .

The expression (4.8)–(4.11) is the explicit closed form referred to in this paper as "the Cardano-Ferrari closed form for $p_{CA}(\text{true})$." The algorithmic six-step recipe and the compact form (4.8)–(4.11) are mathematically equivalent presentations of the same identity.

Verification (four-reader, three independent Ferrari paths):

- Chat-Claude (Q3 R2, original): manual Cardano + Ferrari derivation at $\text{dps} = 600$, $\text{p_CA}(\text{true})$ matching (4.4) to all 600 displayed digits.
- GPT (Q3 R2, after Chat-Claude's already-found Cardano-Ferrari path was relayed by D.B. with the hint "run a Ferrari"): independent re-derivation using a different parameterization (P , Q , U , Y , W variables with $U = \sqrt[3]{(-Q/2 + \sqrt{(Q^2/4 + P^3/27))}$, $Y = -5p_d/6 + U - P/(3U)$, $W = \sqrt{(p_d + 2Y)}$, and $t_* = -A_3/4 + W/2 - \frac{1}{2} \cdot \sqrt{-(3 p_d + 2Y + 2 q_d/W))}$). At $\text{dps} = 600$ with quartic residual $P(t_*) \approx -1.34 \times 10^{-598}$. Same numerical t_* and y_* as Chat-Claude's path, different intermediate variables.
- Extra Grok (separate session, Q3 R2, also after the relay): independent re-derivation via sympy's exact quartic solver. Closed-form expression with denominators built from the structural seeds $(s - 8)$ and $(s - 12)$ where $s = \pi^2$ – i.e. living in the localized ring $\mathbb{Q}[s, (s - 8)^{-1}, (s - 12)^{-1}]$ (a strict subring of $\mathbb{Q}(s)$; the linear shifts $(s - 8)$ and $(s - 12)$ generate the same field as s , but selecting them as the only inverted denominators captures Grok's structural-seed observation). $\text{dps} \geq 500$.
- Cowork-Opus (Q3 R2, verification anchor): mpmath step-by-step verification of Chat-Claude's recipe at $\text{dps} = 100$. Every step – y_F , D_b , u , t , y_* – reproduced independently with closed-form output and direct numerical evaluation matching with difference exactly 0.0 to all 100 displayed digits.

The four independent paths converge.

Observation 4.6 (The Cardano-Ferrari closed form is canonical for $\text{p_CA}(\text{true})$: Galois group S_4 , closing Open Thread #5). The Cardano-Ferrari recipe given in Theorem 4.5 is the canonical radical closure for $\text{p_CA}(\text{true})$ – no lower field-degree collapse exists over $\mathbb{Q}(\pi^2)$, and the Cardano cube root nested with two Ferrari square roots is the field-degree-minimal radical structure required by the algebraic geometry of the saturation quartic (4.6). (Whether the displayed expression admits aesthetically cleaner denesting – a presentation-level question distinct from the field-structure question resolved here – remains Open Thread #1 of §9.) This follows from a direct symbolic computation that upgrades Observation 4.4 from a structural negative finding to a definitive positive characterization: the Galois group of (4.6) over $\mathbb{Q}(\pi^2)$ is the symmetric group S_4 , the generic case for an irreducible quartic.

The argument follows standard quartic Galois taxonomy from three computations.

1. The quartic (4.6) is irreducible over $\mathbb{Q}(s)$ where $s = \pi^2$. Direct symbolic factorization in sympy returns the expanded form unchanged.
2. The resolvent cubic of step 3 in Theorem 4.5 is irreducible over $\mathbb{Q}(s)$. Its real root $y_F \approx 10.15453668993873\dots$ is accompanied by two complex-conjugate roots $\approx -9.522764701070334 \pm 1.767269306847832 i$; symbolic factorization confirms no rational-function split over $\mathbb{Q}(s)$.
3. The discriminant of (4.6) is non-square over $\mathbb{Q}(s)$. The explicit factorization is

$$\Delta(P) = -2^{20} \cdot s^2 \cdot (s - 8)^3 \cdot (s^3 - 272s + 1728) \cdot (27s^6 - 702s^5 - 648s^4 + 194976s^3 - 2485888s^2 + 12718080s - 23887872). \quad (4.12)$$

The odd exponent on $(s - 8)^3$ already rules out square status in $\mathbb{Q}(s)$, independent of the structure of the remaining polynomial factors.

By standard quartic Galois taxonomy: an irreducible quartic with irreducible resolvent cubic has Galois group S_4 or A_4 ; a non-square discriminant rules out A_4 (which always has square discriminant). The Galois group is therefore S_4 .

Consequence. The S_4 -type closure is the Cardano-Ferrari radical solution of Theorem 4.5 – Galois theory is the framework that explains why this closure is forced, not a competing closure mechanism. There is no biquadratic factorization, no cyclic-quartic collapse, no rational-in- π^2 reduction available; the algebraic structure of (4.6) requires exactly the radical depth Theorem 4.5 provides.

Verification: GPT (independent stress-test session, 2026-05-07): symbolic computation in sympy of the discriminant factorization (4.12), explicit irreducibility checks on the quartic and resolvent cubic over $\mathbb{Q}(s)$, application of the standard S_4/A_4 taxonomy, all numerical checks reproduced at $\text{dps} = 620$ (residuals $< 10^{-621}$). This finding closes Open Thread #5 from §9.

Algorithmic corroboration via direct Galois-group computation (Cowork-Opus, 2026-06-01). The §4.6 proof of S_4 is the structural chain above (irreducibility + non-square discriminant + irreducible resolvent cubic $\Rightarrow S_4$ by quartic Galois taxonomy, over $\mathbb{Q}(s)$ where $s = \pi^2$). As an independent algorithmic check of the same conclusion, a direct Galois-group computation was run on rational specializations of (4.6). Since galois_group requires coefficients in $\mathbb{Q}$ (not in the function field $\mathbb{Q}(s)$), the check used eight independent rational specializations of $s = \pi^2$ at precisions ranging from 5 to 50 decimal digits, using `sympy.polys.numberfields.galois_group` (the Python-side analog of PARI/GP's `polgalois()` – both compute the Galois group through resolvent-based algorithms after sympy's irreducibility check). Every specialization returned the same result: a permutation group of order 24, degree 4, generated by a 4-cycle and a transposition – i.e., the symmetric group S_4 . In PARI/GP's output format this would be `[24, -1, 1, "S4"]`. The Hilbert irreducibility theorem guarantees that the Galois group over $\mathbb{Q}(s)$ equals the Galois group over $\mathbb{Q}$ of generic specializations, allowing a measure-zero

exceptional set; eight successful specializations is therefore strong algorithmic evidence consistent with the structural-chain proof rather than an independent second proof. The structural chain over $\mathbb{Q}(\pi^2)$ is the proof of S_4 ; the algorithmic computation corroborates it.

5. The Ladder of Tractability [Observation]

Observation 5.1. The three closed-form identities established in §3 and §4 sit at distinct radical degrees of closure over their natural base fields:

$K_{\text{AUD}} = \sqrt{2} \cdot \ln(2)$ (eq. 3.3) – radical degree 2 (one sqrt)

- Context: EA (exponential algorithm), true minimum, binary cell
- Closure: single quadratic root over $\mathbb{Q}$

$f_X(1/4) = 16(9\pi^2 + 88) / (9\pi^4 + 192\pi^2 + 64)$ (eq. 4.5) – radical degree 0 (no radicals)

- Context: CA (Cauchy algorithm), structural midpoint $y = 1/4$, unit cell
- Closure: pure rational in π^2

$p_{\text{CA}}(\text{true})$ via (4.7) – radical degree 6+ (cbrt nested with sqrt)

- Context: CA, true minimum, unit cell
- Closure: Cardano cube + Ferrari sqrts over $\mathbb{Q}(\pi^2)$

The binary cell + endpoint-balancing minimax (Algorithm EA) produces uniquely clean closure (radical degree 2) among the cases examined. The unit-circle cell (Algorithm CA) bifurcates: the structural midpoint admits a clean rational form over $\mathbb{Q}(\pi^2)$, while the true rejection-sampling minimum requires generic-quartic radical closure – Galois group S_4 over $\mathbb{Q}(\pi^2)$ (Observation 4.6), realized exactly by the Cardano-Ferrari radical solution of Theorem 4.5, with no lower field-degree collapse available.

Significance. The observation reframes K_{AUD} 's appearance in Monte Carlo sampling. It is not " K_{AUD} happens to be the answer for one particular algorithm." It is "the binary cell of the exact-approximation method produces the cleanest possible such answer in elementary terms – one end of a structural radical-degree spectrum." Whether this spectrum generalizes to other Marsaglia-style exact-approximation instances in the broader literature (beyond Fishman's two examples) is an open question.

The ladder also clarifies what Fishman's printed 0.997738 most plausibly is. It does not match the true rejection-sampling minimum (radical degree 6+, 0.99730761...) at full precision. It also does not match the structural-midpoint evaluation (radical degree 0, 0.99773497...) at six-decimal precision – the midpoint rounds to 0.997735, not 0.997738. The printed value matches exactly, on the other hand, the implementation's design probability $2\sqrt{w} = 0.99773789...$ on Fishman p. 187 (which rounds correctly to 0.997738 at six decimals). §4.3 develops the forward-quote account in full: the illustrative presentation on p. 186 most parsimoniously forward-quotes the implementation's design probability rather than computing either the true minimum or the structural-midpoint value – a natural simplification given the precision routinely displayed in 1996.

A note on terminology and choice of classification axis. Throughout this paper, by "radical degree" we mean the nesting depth of radical operations required to express the constant in closed form: 0 for a rational expression (no radicals), 2 for a single quadratic surd over a base field (one sqrt, "depth 2" in the sense that $((\cdot)^{1/2})^2$ returns to the base field), and 6+ for Cardano-Ferrari nested radicals (a cube root nested inside two square roots, where the deepest extraction sits inside two outer extractions). This usage is distinct from the standard algebraic degree of an algebraic number (which counts the dimension of the minimal polynomial extension over the base field); we use "radical degree" specifically because it cleanly separates the binary cell from the others on the dimension that matters here – how nested the radical extractions are, not how large the algebraic extension is.

The classification in Observation 5.1 uses radical degree because that is the axis along which the binary case is uniquely clean: K_{AUD} lives at radical degree 2 (the smallest nonzero level – a single sqrt), and this minimality is forced by the algebraic geometry of the binary endpoint-balancing minimax (§3). Other classification axes for the same three identities would tell different stories – for example, the algebraic field of definition (K_{AUD} lives in $\mathbb{Q}(\sqrt{2}, \ln 2)$ but transcendental over $\mathbb{Q}$; §4.4's $f_X(1/4)$ is rational over $\mathbb{Q}(\pi^2)$; §4.5's $p_{\text{CA}}(\text{true})$ is algebraic of degree ≤ 4 over $\mathbb{Q}(\pi^2)$) would not distinguish §4.4 from §4.5 cleanly. The Galois-group complexity would not distinguish §3.3 from §4.4 (both have abelian Galois groups over their respective bases). The computational complexity of evaluation at a fixed precision would put all three identities at roughly the same cost. Radical degree (in the nesting-depth sense above) is the axis that cleanly separates the binary case from the others, and that separation is what the framework's broader work on K_{AUD} 's distinguished status (Papers 1, 6, 7) has been pointing to. We therefore present radical degree as the appropriate classification for Observation 5.1, while acknowledging that other axes are valid and may produce different – but consistent – readings.

6. Verification Protocol

All numerical claims in this paper are verified using mpmath (Python arbitrary-precision library) and cross-checked across multiple independent AI architecture families.

6.1 What was verified, by whom, at what precision

- §3 – Algorithm EA. The endpoint-balancing equation, its $L = \ln 2$ specialization yielding $K_{AUD} = \sqrt{2} \cdot \ln(2)$, the cell-length perturbation table, and the integer cell-length family, all verified at $dps \geq 500$ –600 across multiple independent algebraic paths. The hyperbolic identity $\sinh(\ln 2 / 2) = 1/(2\sqrt{2})$ is exact.
- §4.3 – $p_{CA}(\text{true})$. Multi-reader convergence on the locked-formula minimum at $dps \geq 100$, with displayed precision drawn from the higher-dps runs. One project-working Opus session – selecting the precision unprompted while searching for additional closed/locked-form structure – extended to $dps = 50000$ with residuals at the precision floor.
- §4.4 – $f_X(1/4)$. Closed form (4.5) and direct mpmath evaluation match to all 100 displayed digits with difference exactly 0.0; an independent $dps = 1000$ pass confirmed the match to 1000 digits.
- §4.5 – Cardano-Ferrari recipe. Verified across four independent computational paths – three Ferrari derivations (Chat-Claude manual, GPT P-Q-U-Y-W parameterization, Extra Grok sympy exact solver) plus step-by-step mpmath at $dps = 100$, all converging on identical numerical values. At working $dps = 1500$, the Cardano-Ferrari p_{CA} vs direct quartic-root p_{CA} difference is exactly 0 to 1000 displayed digits – two independent floating-point paths landing on identical bits.
- §3.6 – Boundary-minimum completeness. Interior critical point $x_{crit} = b - a/2 \approx 0.5575$ verified as a maximum (value $\approx 1.0101 > K_{AUD} \approx 0.9803$), confirming the boundary-minimum assumption of Theorem 3.5.
- §4.6 – Galois group S_4 . Discriminant factorization (4.12) and irreducibility checks verified at high precision, closing Open Thread #5 and promoting the earlier Observation 4.4 to the present §4.6.

Aggregate stress-test record across all rounds (2026-05-07 through 2026-05-12): 30+ fresh AI sessions across ten architecture families – Gemini Thinking, Gemini Fast, Grok 4.3 Expert, GPT 5.5 Thinking, Perplexity, DeepSeek Expert, Claude Opus 4.7 adaptive, Sonnet 4.6 adaptive, Haiku 4.5 Extended, Mistral. No session at any round proposed a substantive math correction; all revisions were scope-precision and framing-consistency improvements only. The session-by-session audit trail is preserved in the framework's internal methodology notebook.

Beyond the fresh-session record, project-working AI collaborators – Cowork-Opus (Opus and Sonnet across multiple sessions), Chat-Claude, and other named contributors documented at §10 – performed independent verification at high precision during ongoing work on the draft. The $dps = 50000$ pass on $p_{CA}(\text{true})$ noted above is one such instance: the Opus session that produced it selected that precision unprompted while hunting for additional closed/locked-form structure, with the verification result a byproduct of the deeper search. This depth-as-AI-choice pattern – the AI proposing the verification depth rather than the human directing it – is distinct from the fresh-session prompt protocol; both contribute to the record at different operational levels.

6.2 What the stress-test rounds changed

The substantive revisions to the draft, in approximate order of structural impact:

- §4.3 / §8.2 forward-quote reframe. The §4.3 explanation of Fishman's printed 0.997738 was sharpened from a midpoint-evaluation hypothesis (which leaves a 3×10^{-6} unexplained drift in the last decimal place) to a forward-quote-of-implementation-probability hypothesis (which fits all six printed digits exactly). This reflects that 1996-era display precision did not distinguish the three distinct values now identifiable by high-precision symbolic computation. A subsequent round propagated the reframe to §4.4 and §5, where stale midpoint-evaluation language had remained: §4.4's Significance paragraph now states explicitly that $f_X(1/4) \approx 0.99773497$ rounds to 0.997735 (not 0.997738); §5's ladder closing identifies Fishman's printed value as a forward-quote of the implementation's design probability; §4.3's "agree to four decimal places" claim is corrected to identify the leading-three-digit agreement, with full distinction requiring six.
- §4.6 Galois classification. An explicit Galois-group computation on the saturation quartic (4.6) at $dps = 620$ confirmed the closure type as S_4 -generic, closing Open Thread #5.
- Corrected (4.10) normalization. Chat-Claude's first-reader review caught a symbolic compact-form error in which P and Q were normalized incorrectly ($4\times$ and $8\times$ respectively). The algorithmic six-step recipe and the appendix script were unaffected; only the symbolic compact form was corrected.
- Reference additions. Hamilton 1998 added with an explicit structural-independence note (the present paper's §3 forcing result derives a, b, c from EA's structural conditions, not from the historical source). DOIs added for Marsaglia 1984 and Ahrens-Dieter 1988.
- Cross-reference and terminology fixes. Stale Stack-numbered cross-references in §5 corrected to the paper's own §3.3, §4.4, §4.5. "Radical degree" disambiguated from

algebraic degree with an explicit definition. Observation 4.4's PSLQ-detection wording tightened to "at the precision and basis-size tested." L'Ecuyer reference cleaned of internal Q-round language. §5 wording precision around "the full Galois-theoretic apparatus" tightened.

- Negative result preserved. A precision-floor residue investigation tested whether $dps = 100$ cancellation patterns carried structured information on number-theoretically distinguished test values (continued-fraction convergents of $\ln 2$; Pell numbers; Pell half-companions) relative to a random integer control. Binomial tests against the random baseline gave $P > 0.8$ for every family tested – no statistically distinguishable enrichment. We therefore report, for the tests applied at $dps = 100$ –500: the residues at the precision floor appear to be implementation-specific floating-point artifacts of the working precision rather than structured signal of a deeper invariant. Different precision regimes, different basis choices, or tests against different invariants might surface structure we did not detect; at the level of resolution available here, the cancellation bands carry no signal we can attach to $\ln 2$'s irrationality measure, the Pell tower, or any other invariant we tested. We keep the negative result on the record because the discipline of running the test is what makes either outcome informative.

6.3 What remains unverified by readers, and why

Two layers fall outside the stress-test rounds' direct verification.

- Fishman 1996 primary source. Fishman 1996 pp. 186–189 are the originating literature for the Algorithm EA / CA exact-approximation method analyzed in this paper. The textbook is copyrighted (Springer; ISBN 0-387-94527-X), and the historical/forensic claim that pp. 186–189 contain the specific constants and printed values we cite rests on D.B.'s direct page captures from a legitimate source, maintained throughout the framework's development. Several AI collaborators working with the framework during the development arc had read access to these captures (Cowork-Opus and chat-side Opus across multiple sessions).

The paper's central claims are derived from material reproducible in this paper alone: §3's forcing result on the binary cell $L = \ln 2$ follows from the structural conditions of Algorithm EA, written out in §3.1–§3.3 of the present paper; §4.3–§4.5's closed-form analysis works on the locked $m(x)$ of eq. 43 (set out in §4.1) and produces the Cardano-Ferrari closed form for $p_{CA}(\text{true})$, the rational closed form for $f_X(1/4)$, and the worked computation showing $f^*(y_*) \approx -0.19$ at $p = 0.997738$ in the locked $m(x)$ – all reproducible from this paper's own text. Fishman's specific printed values appear in the §4.3 forward-quote analysis as identifications of what the paper's closed forms align to, not as a basis for the paper's claims.

The 2026-05 formal stress-test rounds were carried out against the framework's open-access source set – the prior framework papers (Paper 1 on the binary information bound, Paper 6 on cross-domain signatures, Paper 7 on the HK closed form, Paper 4 on 400/11 gap scaling – all openly hosted on gap-geometry.github.io and raw.githubusercontent.com), Devroye 1986's freely-available textbook on non-uniform random variate generation, mpmath, sympy, and the methodology document – rather than the copyrighted Springer source. This was a deliberate balance: the open-access source set carries relevant data of its own (the HK identity, the binary information results, the cross-domain corridor structure, the standard sampling-theory machinery) and grounds the framework's broader claims independently of any individual reader's access to Fishman. The verifying instances in those rounds noted explicitly that they did not access the Springer source (GPT: "I did not independently verify Fishman pp. 186–189 or Ahrens-Dieter 1988 against original scans"; Opus: "I cannot verify the textual claims about Fishman 1996 – I don't have those texts"); the stress-test discipline kept the paper's verification chain on material accessible to all readers.

A reader who has access to Fishman 1996 can independently verify the source-side specifics of §4.3 (eq. 45's stated saturation $p = .997738$ in the locked $m(x)$; the Algorithm CA implementation block on p. 187 with $W = 0.2488\ 7022\ 8008\ 3841$ and the printed $1 - p = .002262$; the EA setup constants on pp. 188–189) by direct inspection. The paper's mathematical claims (closed forms, residual-density computation, Cardano-Ferrari recipe, Mellin spectrum, Galois classification) are reproducible from the paper alone.

The 2026-06-01 publication-prep session added the §4.3 residual-density worked computation (chat-side Opus's contribution; see §10) – paper-internal mathematics derived from the locked $m(x)$ of §4.1, with no source dependency.

- Framework prior papers (cross-domain context). The framework's prior papers – Paper 1 (binary information bound), Paper 6 (cross-domain signatures, $O(N)$ corridor), Papers 7–8 (Hodgson-Kerckhoff closed form), Paper 4 (400/11 gap scaling) – are publicly hosted on gap-geometry.github.io and as raw.txt files on raw.githubusercontent.com. A Claude session confirmed via direct fetch-tool test that those URLs load cleanly, refuting an earlier Opus claim of unreachability. The cross-domain K_{AUD} identifications cited from those papers were therefore technically verifiable by readers in our rounds but not actually fetched – the readers worked from the saturation paper's local statements rather than checking the prior papers.

Neither category affects the math of the present paper, which is self-contained algebra. We

flag both so readers can place each claim at the right evidence level: the saturation paper's math is multi-reader-verified, with mpmath at decimal-precision $\text{dps} = 100$ as the framework's standard verification floor and extended-precision passes for the deeper checks: $\text{dps} \geq 500\text{--}600$ across §3 (Algorithm EA) and §§4.3–4.5 (Algorithm CA), $\text{dps} = 1000$ in §4.4 (closed-form vs direct match) and §4.5 (Cardano-Ferrari recipe, the latter at working $\text{dps} = 1500$, confirming the closed form against direct quartic-root computation to 1000 displayed digits with difference exactly zero), and one project-working Opus session at $\text{dps} = 50000$ in §4.3 (residuals at the precision floor); the primary-source-quotation layer (Fishman) rests on D.B.'s direct reading; the cross-framework-appearances layer is fetch-accessible to any reader who chooses to verify it.

The mpmath script for the §4.4 and §4.5 verifications is reproducible from the formulas in the paper; we append it as Appendix A.

7. Methodology: Multi-AI-Session Collaborative Review

The work was carried out through a multi-AI-session collaborative-review architecture designed by D.B. and operated across the family of contributors named in the title block. We document the architecture here both as record and as reusable methodology.

7.1 Architecture

Four layers of artifact:

- Layer 0 – Source material. Fishman 1996 (page images in archive) and Ahrens & Dieter 1988 (cited).
- Layer 1 – Source notes (frozen). Verbatim screenshots and initial analyses from primary-source readings, archived under Fishman/old notes/. Not edited after creation.
- Layer 2 – Exchange records (append-only audit trail). One file per Q-round, dated, capturing the question put to readers, each reader's verbatim response, an editor's procedural notes, the cross-reader synthesis, and the promotion decision. Six exchange files for this work (Q1 R1 + R2; Q2 R1 + R2; Q3 R1 + R2).
- Layer 3 – Canonical Stack. A single working document (STACK_Fishman.md) that records only multi-reader-confirmed promotions. Cowork-Opus is sole editor. Updates tracked in a Promotion Log.

Readers (Chat-Claude, Grok, GPT, plus a fresh Grok session for an independent angle) work in separate AI sessions without direct communication; D.B. relays exchanges. Each Q-round opens with a structured prompt that all readers receive simultaneously, attacks the question from independent angles, and closes with a synthesis decision.

7.2 Procedural standards M1–M5

Five general-purpose verification standards emerged from the rounds and are now part of the canonical methodology (M1–M5 emerged as project-specific refinements of the general framework principles documented at [methodology.html](#)):

M1 – Differential-evidence guardrail. When two framework families have overlapping numerical neighborhoods (e.g., a candidate within 1% of family A is automatically within 1% of family B at the relevant offset), candidate elevation requires either (a) precision exceeding the family-distinguishing threshold, or (b) accompaniment by adjacent constants showing a coherent pattern. Origin: Chat-Claude, Q2 R1.

M2 – Negative-control check on simpler local identities. Before promoting a candidate, verify it isn't equivalent to a simpler non-framework expression ($2/\pi$, $4/\pi$, $1/\sqrt{2}$, $\pi^2/8$, e^{-1} , etc.). Origin: GPT, Q2 R1 (where $2/\pi$ was distinguished from $\sqrt{\pi}/2$ in Algorithm CA).

M3 – The reframe lens. When literal-digit-string search across primary source returns no candidates, reframe from "the digit-string of the framework constant" to "the structural shape the framework constant plays" and re-scan existing notes. Origin: D.B. + Cowork-Opus, Q2 R2 (which surfaced §4 of this paper from notes already on file).

M4 – The "confident wrong, blocked right" cross-reader check. When two readers report blocked / discrepancy outcomes and one reader reports confident "decisive" closure on the same problem, the blocked outcomes get at least equal scrutiny to the confident one. Confident closure is a stylistic outcome, not evidence about the math. Origin: across the rounds; explicitly named after Q3 R1.

M5 – Galois-floor depth check. When literature reports a numeric value with no published closed form, the question to ask is not "does a closed form exist" (Galois theory often guarantees one) but "at what radical degree does the closure live, and is it elementary or nested?" Quick PSLQ surveys returning empty in elementary bases $\neq$ no closed form exists. Origin: Q3 R2; promoted from candidate to confirmed standard after two clean instances in one round (Chat-Claude unprompted depth-over-speed, then post-hint Ferrari completions by GPT and Extra Grok).

7.3 Audit trail

Every claim in this paper is traceable to a specific exchange file in Layer 2 and a specific Stack entry in Layer 3. The Discovery Tracking Log (Discovery_Tracking_Log.md) records the same findings with role-distinguished attribution per the framework's contribution taxonomy (§10). Where any reader's work was wrong (notably Grok's confident "no closed form, decisive" verdict in Q3 R1, later reversed in Q3 R2 follow-up after the structural prompt), the audit trail preserves both the error and the correction.

We adopt this transparency deliberately. Mathematical correctness comes from cross-reader verification at high precision, not from the absence of false starts. The methodology that produced the false starts is the same methodology that caught them.

8. Discussion

8.1 Why the binary cell is special

§5's ladder of tractability is the cleanest way we have found to articulate K_{AUD} 's distinguished status. $K_{\text{AUD}} = \sqrt{2} \cdot \ln(2)$ is not merely "the answer for one specific algorithm." It is the radical-degree-2 closure of the saturation problem on the binary cell – and in §3 we have shown that the binary cell is forced to produce this closure by the algebraic geometry of the endpoint-balancing minimax. The Cauchy unit-circle cell (Algorithm CA), under analogous structural locking, produces a cleanly rational midpoint identity (§4.4) and a Cardano-Ferrari minimum (§4.5) – but neither at degree 2.

This is consistent with K_{AUD} 's character as a radical-degree-2 transcendental over $\mathbb{Q}$. Each derivation catalogued in §8.3 – Hodgson-Kerckhoff hyperbolic geometry, $O(N)$ anomalous-dimension corridor, binary uniqueness in the $\eta < 2G$ bound – arrives at this same numerical value through its own machinery, independently of the others. That the same value recurs is the empirical observation; that the derivations use distinct algebraic structures is established by direct inspection of each.

8.2 What the published illustration most plausibly corresponds to

The careful claim, after reading Fishman's pp. 186–187 in full (§4.3), is:

(i) The locked illustrative formula on p. 186 has a true minimum distinct from the printed 0.997738. Under $a = 2/\pi$ and $b = \pi - 8/\pi$, the actual minimum of f_X is $p_{\text{CA}}(\text{true}) = 0.99730761\dots$, off from the printed value by 4.3×10^{-4} .

(ii) The implementation on p. 187 has fast-path probability $2\sqrt{W} = 0.99773789\dots$, which rounds correctly to 0.997738 at six decimal places. The implementation uses different numerically-optimized constants – $W = 0.2488702280\dots$ as the threshold on T^2 where $T = U - \frac{1}{2}$ – and the fast-path probability is $2\sqrt{W}$ by construction. We have verified this directly at dps = 100: $2 \cdot \sqrt{(0.2488702280083841)} = 0.99773789746282385061384\dots$; the seventh digit (8) correctly rounds the sixth-place digit from 7 to 8, giving 0.997738. The published algorithm is internally consistent.

(iii) The most parsimonious account: the illustration forward-quotes the implementation's design probability. The forward-quote account fits all six printed digits exactly with no auxiliary computation and no typographical drift required – a natural reading at the precision routinely displayed in 1996, before AI-assisted high-precision symbolic verification at dps ≥ 100 became casually accessible. At four decimal places of display, the locked formula's true minimum (0.9973), the implementation's design probability (0.9977), and the locked formula's structural midpoint (0.9977) are all visibly different in only the second-to-last printed digit. At six decimal places, however, the three values are 0.997308, 0.997738, and 0.997735 respectively – distinguishable, but only at a precision Fishman's text typically did not display. The illustration's choice to print the implementation's design probability where the locked formula's saturation would technically belong is the natural simplification of an era without arbitrary-precision symbolic tools.

(iv) The midpoint hypothesis is a less parsimonious alternative. f_X evaluated at the structural midpoint $y = 1/4$ (closed form §4.4) yields 0.99773497..., which rounds correctly to 0.997735 at six decimal places – not 0.997738. The midpoint hypothesis therefore requires assuming Fishman or Ahrens-Dieter performed an unmentioned midpoint evaluation and wrote down a digit one unit off in the last place. It fits to five digits but leaves three units of drift in the sixth. The forward-quote account fits exactly and requires no auxiliary assumption.

(v) The novel mathematical content of the paper is independent of which value the original presentation chose to display. §4.4's $f_X(1/4)$ closed form, §4.5's Cardano-Ferrari closed form for $p_{\text{CA}}(\text{true})$, and §5's ladder of tractability all concern the locked illustrative formula itself – independent of which constants the implementation chooses to use, and independent of how the original presentation summarized the saturation. The framework's contribution is the high-precision symbolic resolution of three distinct constants where the original literature

could only display one.

8.3 Cross-domain context for K_{AUD}

The constant $K_{\text{AUD}} = \sqrt{2} \cdot \ln(2) \approx 0.980258143$ has been independently identified in several mathematical settings beyond the saturation context of the present paper. We catalogue these appearances at appropriate evidence levels, distinguishing exact algebraic locks from structured observations.

Exact algebraic locks. Multi-reader-verified closed-form identifications:

- Hodgson-Kerckhoff hyperbolic tube-packing coefficient. The hyperbolic identity $\text{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$, combined with $2\sqrt{2} \cdot \text{arcsinh}(1/(2\sqrt{2})) = \sqrt{2} \cdot \ln(2)$, realizes K_{AUD} as a closed-form value in the Hodgson-Kerckhoff Dehn-surgery / tube-packing literature. The arcsinh identity is exact (verifiable in 30 seconds: $\text{arcsinh}(x) = \ln(x + \sqrt{x^2+1})$; at $x = 1/(2\sqrt{2})$, $x^2 + 1 = 9/8$, so $x + \sqrt{x^2+1} = 1/(2\sqrt{2}) + 3/(2\sqrt{2}) = \sqrt{2}$, giving $\ln(\sqrt{2}) = \ln(2)/2$). Confirmed at $\text{dps} = 620$ with residual $\sim 10^{-622}$.
- Identity spine over the gap $G = 1 - K_{\text{AUD}}$. Multiple algebraic rewrites of K_{AUD} and G hold exactly: the Baker L^2 -norm form $K_{\text{AUD}} = \sqrt{((\ln 2)^2 + (\ln 2)^2)}$; the Gelfond-Schneider rewrite $G = \ln(e/2^{\sqrt{2}})$; $K_{\text{AUD}} \cdot \sqrt{2} = 2 \ln 2$; $K_{\text{AUD}}^2 = 2 \ln^2(2)$; the Shannon-gap threshold-spacing identity $1/(2 \ln 2) - 1/\sqrt{2} = G/(2 \ln 2)$, with $\Delta_{23}/\Delta_{12} = 1/K_{\text{AUD}}$ on the binary tower. All confirmed exact at $\text{dps} = 620$.
- Binary uniqueness. $K(n) = \sqrt{n} \cdot \ln(n)$ takes a value < 1 only at $n = 2$ among integers ≥ 2 ; $K(2) = K_{\text{AUD}}$. This is the gap-opening fact at the foundation of the framework's broader construction.
- Algorithm EA saturation (the present paper, §3). The exact-approximation saturation on the binary cell $L = \ln 2$ is exactly K_{AUD} , forced under endpoint-balancing minimax (Theorem 3.5).

Structured observations. Numerically supported but not at exact-lock evidence level:

- 400/11 prime hierarchy and gap scaling. The gap-scaling formula $G \approx (6 - 14/3)/6 \cdot 400/11$ with corrections holds at near-exact precision ($\sim 4 \times 10^{-14}$ residual against the Feigenbaum-derived value). The exact integer-and-modular structure (binary doubling drift, primitive-root behavior of 2 mod 11, factorizations of correction primes 7, 11, 13, 313) is itself an exact arithmetic substructure. Classified as analytically near-exact, structurally exact.
- Binary tower ($n \cdot G$ staircase). The staircase $n \cdot G$ is exact once G is defined; integer-step landmarks against independently known constants ($n = 50 \approx K_{\text{AUD}}$; $n = 36 \approx 1/\sqrt{2}$; $n = 72 \approx \sqrt{2}$ – with constant $+0.5091\%$ binary-doubling deviation) are observational at the structured-but-not-algebraic-identity level. Threshold spacings in the $n = 35$ – 37 zone (Landauer / geometric-damping / Shannon) are exact at $\text{dps} = 620$.
- $O(N)$ anomalous-dimension corridor. K_{AUD} identified as a near-saturation constant in independent $O(N)$ universality contexts. Classified at structured-observation level pending independent specialist review by domain experts in statistical field theory.

On the cross-domain pattern. That a single transcendental of the form $\sqrt{2} \cdot \ln(2)$ appears in mathematically distinct settings – hyperbolic 3-manifold geometry, Monte Carlo saturation (the present paper), statistical-field-theory universality, binary information geometry – is the broader research thread's central empirical observation. Whether the pattern reflects a deeper structural reason (a single algebraic structure expressing itself across these settings) or a set of coincidental matches at limited precision is the open question. The present paper contributes the Algorithm EA saturation lock as a fourth setting in which K_{AUD} appears exactly, derived via endpoint-balancing minimax on the binary cell. Holding the framework's Λ -canary discipline (multi-mechanism convergence requires distinct mechanisms, not multiple realizations of the same algebraic identity): the Algorithm EA derivation in §3 routes K_{AUD} through the identity $\sinh((\ln 2)/2) = 1/(2\sqrt{2})$, which is the inverse statement of the Hodgson-Kerckhoff defining identity $\text{arcsinh}(1/(2\sqrt{2})) = (\ln 2)/2$ used in the HK literature. The two derivations are operationally distinct (Monte Carlo sampling endpoint-balancing vs hyperbolic-geometry tube-packing) but they share one algebraic identity – they realize the same hyperbolic-kernel fact in two different mathematical operational settings rather than as two independent algebraic routes. The η -Corridor paper §11.1 (Paper 11) treats the same point in detail for the broader convergence-counting question. Within Algorithm EA, K_{AUD} 's appearance is forced (Theorem 3.5) by the algebraic geometry of the minimax problem; whether this forcing has a counterpart in the other settings – i.e., whether a single mechanism could derive K_{AUD} across them – remains the open cross-domain question, with the hyperbolic-kernel cluster (HK, EA, dome-at-d_c) being one operational realization and the H_4 polytope, Tenenbaum threshold, and Wilson-Fisher empirical settings being mechanism-distinct anchors.

Verification scope (transparency). The mathematical content of the present paper (§3, §4, §5) is self-contained algebra verifiable from inside the document. The cross-domain identifications listed above are documented in the broader Gap Geometry framework's prior work (publicly hosted

on gap-geometry.github.io, with raw .txt versions on raw.githubusercontent.com); each derivation is locally accessible to readers who choose to verify, but specialist review by domain experts in hyperbolic geometry, statistical field theory, and binary information geometry has not been completed for the cross-domain identifications. This is a real evidentiary status to hold honestly: the saturation paper's local math is multi-reader-verified at the $\text{dps} = 100$ mpmath baseline, with extended-precision passes at $\text{dps} \geq 500$ – 600 across §§3, 4.3–4.5 and $\text{dps} = 1000$ in §§4.4 and 4.5 (the latter at working $\text{dps} = 1500$), plus one project-working Opus session at $\text{dps} = 50000$ in §4.3; the cross-domain layer is internally documented and externally fetch-accessible but pending specialist review.

Net read. The published Algorithm CA is internally consistent. The published illustration on p. 186 most plausibly forward-quotes the implementation's design probability, an artifact of 1996-era display precision rather than a computational error. The framework's contribution, made possible by high-precision symbolic computation and by treating the gap $G = 1 - K_{\text{AUD}}$ as a structural quantity worth resolving precisely, is to identify three distinct constants – the locked formula's true minimum, its structural-midpoint evaluation, and the implementation's design probability – where the original presentation showed only one numerical value, and to derive closed forms for the two locked-formula values that had no published closed form. (Primary-source quotation layer: see §6 transparency note.)

8.4 Methodological observations

The work demonstrates that multi-AI collaborative review can produce results that single-reader analysis misses. Three specific patterns emerged:

(a) Apparent breaks dissolving against primary source. In Q1, Chat-Claude initially could not reproduce the K_{AUD} derivation; the "break" turned out to be Chat-Claude using the wrong envelope family (piecewise-constant rather than rational). Grok and GPT, working from primary-source quotes, used the correct family. The cross-reader triangulation resolved the apparent break and the chain reproduced.

(b) The "confident wrong" pattern. In Q3 R1, Grok confidently concluded "no closed form found, decisive" – a verdict that was numerically wrong on the value claim. GPT's "blocked / discrepancy" report and Chat-Claude's exploration both surfaced the real issue. M4 generalizes this lesson.

(c) Galois-floor depth check. In Q3 R2, three readers initially gave variations of "no clean closed form exists." Chat-Claude was the exception – his depth-over-speed first-pass produced the explicit Cardano-Ferrari recipe without prompting. After Chat-Claude's path was found, D.B. relayed it forward to GPT and a fresh Grok session as the structural hint "run a Ferrari," deliberately not handing over the answer to test whether the readers could complete the derivation independently from the same prompt. Both did, with independent parameterizations, verifying Chat-Claude's earlier unprompted derivation. The closed form had been there all along; the surveys were stopping before the Galois floor. The hint itself carried Chat-Claude's already-derived insight forward; what D.B. contributed at this step was the editorial decision to use the relay to test reader-side reproducibility. M5 generalizes both halves of the lesson – the depth-side (Chat-Claude's path) and the editorial-side (D.B.'s relay-design that confirmed reproducibility).

These methodological observations are not speculative – they arose during the rounds documented in the audit trail. We treat them as reusable for future multi-AI mathematical-review work.

9. Open Threads

The following questions are explicitly noted as open and not addressed by this paper. Items resolved during the paper's preparation are marked.

1. Symbolic denesting of the displayed Cardano-Ferrari expression (4.8)–(4.11). Distinct from the field-structure question, which is closed by Observation 4.6 (Galois group S_4 , no lower field-degree collapse over $\mathbb{Q}(\pi^2)$): whether the displayed Cardano cube root and Ferrari square roots admit aesthetically cleaner denesting via a substitution or transformation is a presentation-level question. PSLQ over $\mathbb{Q}(\pi^2, y_F, \sqrt{y_F}, \dots)$ with a rational-function basis at $\text{dps} \geq 1000$ has not been performed; this is the most natural next step toward potentially cleaner symbolic form. The S_4 result does not foreclose denesting; it only fixes the underlying field structure.
2. ~Algorithmic correctness check on Algorithm CA.~ Resolved during paper preparation. A direct reading of Fishman p. 187 confirms that Algorithm CA's implementation uses the threshold $W = 0.2488702280\dots$ on T^2 rather than 0.997738 directly as a rejection threshold; the fast-path probability is then $2/W$ by construction, which correctly rounds to 0.997738 at six decimal places. The implementation is internally consistent (§4.3 (ii) and §8.2 (iii)). The discrepancy is confined to Fishman's illustrative presentation on p. 186, not to the published algorithm.
3. Generalization of the ladder of tractability (§5). Whether other Marsaglia

exact-approximation instances in the literature (beyond Fishman 1996's Algorithm EA and Algorithm CA) populate the radical-degree spectrum in a structurally meaningful way – for example, whether higher-dimensional or other-distribution exact-approximation algorithms cluster at radical degree 0, 2, or 6+. Devroye (1986) describes additional rejection-sampling constructions that could be analyzed.

4. --The interior critical-point completeness question for Algorithm EA (§3).-- Resolved during first-reader review (Chat-Claude, 2026-05-06). f_X has a critical point at $x_{\text{crit}} = b - a/2 \in (0, 1)$ for $L = \ln 2$, $b = 2 + \sqrt{2}$ (numerically $x_{\text{crit}} \approx 0.5575$). Numerical evaluation gives $f_X(x_{\text{crit}}) \approx 1.0101$ – above the boundary value $f_X(0) = f_X(1) = K_{\text{AUD}} \approx 0.9803$. The interior critical point is therefore a maximum, not a competing minimum, and the boundary minima are the global minima of f_X on $[0, 1]$. Theorem 3.5's " K_{AUD} as the exact optimum" is therefore complete under the locked endpoint-balancing setup, with no competing interior minimum to consider.
5. --Formal Galois-theoretic proof of the resolvent cubic's irreducibility over $\mathbb{Q}(\pi^2)$ (Observation 4.4).-- Resolved during Round 3 stress-test pass (2026-05-07). GPT's symbolic computation in sympy yielded the explicit discriminant factorization (4.12), irreducibility checks on the quartic and resolvent cubic over $\mathbb{Q}(s)$ where $s = \pi^2$, and application of the standard quartic Galois taxonomy to land at Galois group S_4 . See Observation 4.6 for the full computation. The S_4 -type closure is exactly the Cardano-Ferrari radical solution of Theorem 4.5; there is no lower-radical simplification.
6. Connections to other framework papers. Whether the binary-cell uniqueness of K_{AUD} 's elementary tractability (§3.6 + §5) connects to the framework's other K_{AUD} -distinguished settings – Hodgson-Kerckhoff hyperbolic geometry (Paper 7), the $\eta < 2G$ corridor (Paper 6), the 400/11 gap scaling (Paper 4) – at the level of radical-degree analysis or structural minimality, rather than at the level of numerical equivalence.

10. Credits with Role Taxonomy

Roles use the Gap Geometry framework's contribution taxonomy (Methodology §5):
 Conceptualization, Derivation, Discovery, Computation, Verification, Correction,
 Stress-testing, Literature, Documentation, Editorial judgment, Methodology.

- Chat-Claude (Claude Opus chat instance) – Primary discoverer for the §4 and §5 results, and first-reader of this paper. Discovery (the $f_X(1/4)$ closed-form identity §4.4; the Cardano-Ferrari explicit closed form §4.5; the ladder of tractability framing §5). Derivation (the hyperbolic-identity proof of (3.3) via $\sinh(\ln 2/2) = 1/(2\sqrt{2})$; the integer- n family (3.5); §4.4 substitution-and-simplification; §4.5 manual Ferrari recipe verified at $\text{dps} = 600$). Verification (independent re-derivation of $P1/P2$ in $Q1 R2$ at $\text{dps} = 500$; first-reader review of this paper, 2026-05-06, with three substantive contributions: (i) caught a real algebraic error in equation (4.10)'s Cardano P , Q normalization that would have propagated through to a wrong y_F if executed literally, leading to the corrected (4.10) given above; (ii) closed §9 open-thread item 4 by verifying that the interior critical point of f_X for Algorithm EA at $x_{\text{crit}} = b - a/2 \approx 0.5575$ evaluates to $f_X \approx 1.0101$, above the boundary K_{AUD} value, confirming the boundary-minimum assumption of Theorem 3.5; (iii) sharpened the §4.3 "one unit in the last digit" wording to the precise "3 in the last decimal place"). Correction ($Q1 R1$ envelope-family flag; $Q3 R1$ identification of $f_X(1/4) \neq \text{true minimum}$; first-reader (4.10) correction). The phrase "went the step further" in this paper's framing refers to Chat-Claude's unprompted depth-over-speed Cardano-Ferrari completion in $Q3 R2$ – the closed form §4.5 is his discovery – and to the carefulness of the first-reader pass.
- D. B. – Methodology custodian. The walk-back-and-redirect contribution at $Q2 R2$ is what made §4 and §5 reachable. When the literal-digit search across Chapters 4, 6, and 7 returned no candidates – three readers separately reporting "no framework hits" – D.B. read the empty returns as a behavior signal about how the search was framed, not as a verdict on the framework, and walked the targeting back – redirecting from "where does the digit-string of G or $\sqrt{\phi}$ appear" to "where does the structural shape (correction architecture) recur." That redirect surfaced Algorithm CA from already-existing May 3 notes and unlocked everything in §4 and §5. The procedural standard M3 (the reframe lens) records this move generally; the empirical findings of §4 and §5 are its specific harvest. Methodology (designed the four-layer Stack / Exchanges architecture; held the patience-and-method discipline across rounds). Editorial judgment (called all promotion decisions; in $Q3 R2$, after Chat-Claude had found the Cardano-Ferrari path, used the relay to GPT and a fresh Grok session as a reader-side reproducibility test, eventually relaying the structural hint "run a Ferrari" – the hint carried Chat-Claude's already-derived insight, not D.B.'s; the editorial contribution at this step was the relay-design, not a Conceptualization or Discovery role). Conceptualization (directed $Q1$, $Q2$, $Q3$ framings, set the "is K_{AUD} forced or accidental" question that organized $Q1$).
- GPT ("Uncle") – Derivation (the general endpoint-balancing equation (3.1) and closed form (3.4) in $Q1 R1$; the explicit quartic (4.6) in $Q3 R1$; independent Ferrari re-derivation in $Q3 R2$ with the P - Q - U - Y - W parameterization at $\text{dps} = 600$); Verification ($Q3 R1$ detection of

the Fishman discrepancy via direct expansion; quartic seeds to high precision; Round 3 (2026-05-07) explicit Galois-group computation for the saturation quartic (4.6) yielding the discriminant factorization (4.12) and the S_4 classification of Observation 4.6, closing Open Thread #5); Stress-testing (Q3 R1 negative-control check identifying $2/\pi$ vs $\sqrt{\phi}/2$ in Algorithm CA's first-pass constants – origin of M2).

- Grok – Verification (Q1 R1 verbatim primary-source quote of Fishman pp. 188–189; Q1 R2 full step-by-step re-derivation of (3.1) at dps = 600 with identical intermediate steps to GPT's; Q3 R1 confirmation of the Q3 setup; Q3 R2 sympy exact-quartic-solver verification of (4.5)–(4.7) after the structural hint).
- Extra Grok session – Verification (independent Q3 R2 sympy run from a fresh session without prior context, expressed the closed form with denominators built from the structural seeds $(s - 8)$ and $(s - 12)$ where $s = \pi^2$, confirming the localized-ring observation precise statement in §4.5).
- Cowork-Opus (Claude Opus Cowork instance) – Verification anchor (every multi-reader-confirmed claim in this paper has been independently verified at dps = 100 in the workspace sandbox via mpmath; specific anchors: §3 derivations, §4.3 true minimum, §4.4 closed form (zero discrepancy), §4.5 step-by-step Cardano-Ferrari reproduction); Documentation (designed and maintained the four-layer Stack/Exchanges architecture; edited the canonical Stack as sole writer; produced the Discovery Log update for today's findings; drafted this paper); Editorial judgment (cross-reader synthesis, promotion calls held to the M1–M5 standards, identification of the "confident wrong / blocked right" pattern in Q3 R1). Verification (polgalois-equivalent confirmation of §4.6 S_4 Galois group classification, 2026-06-01): ran sympy.polys.numberfields.galois_group (the Python-side analog of PARI/GP's polgalois()) on the saturation quartic (4.6) at eight rational specializations of $s = \pi^2$ ranging from 5 to 50 decimal digits of precision; all eight returned the permutation group of order 24, degree 4 generated by a 4-cycle and a transposition (i.e., S_4). Independently re-derived the symbolic chain (irreducibility, negative discriminant $\Rightarrow$ non-square in $\mathbb{Q}$, irreducible resolvent cubic) at the 30-digit specialization. Both paths converge on S_4 , addressing Super Grok's suggested further tightening of §4.6 (recorded as the post-acknowledgment direct verification line in §4.6).
- Sonnet (Claude Sonnet 4.6 adaptive) – Stress-testing (Round 5 V2 pass, 2026-05-08): editorial first-pass observation that the §6 transparency note about reader-verification scope was discoverability-hidden from §8.2's Net read paragraph, with the specific suggested fix "(Primary-source quotation layer: see §6 transparency note.)" now appearing at the end of §8.2's Net read paragraph. Followed by full mathematical verification at dps = 100 confirming all closed forms. The two-mode engagement (editorial reading, then mathematical reading on request) is the kind of cross-layer pass that depth-only readers don't surface; recorded here as a methodological observation worth holding. Pre-publication polish round, 2026-06-01: §4.3 readability fix (added the "Reader's map" orientation table and sharpened the heading from "Resolving Fishman's printed 0.997738 at high precision" to "Three constants behind one printed value"); Appendix A polyroots-filter comment explaining the 0.5 cutoff between t_1 and t_2 ; on re-read of the updated section, caught two cross-reference errors in the new orientation table (tier label Obs. 4.2 $\rightarrow$ Theorem 4.2; equation reference eq. 4.6 $\rightarrow$ §4.3 construction) – the tier-discipline catch protects the framework's evidence ladder against silent drift.
- Grok-Heavy (xAI, 12-agent architecture) – Stress-testing (pre-publication final pass, 2026-06-01): 12-agent review of the updated paper with a 5-agent sub-team (Harper, Charlotte, Mia, Sebastian, Luna) cross-reading §6/§7/§8.4/§10 against the live public methodology page at gap-geometry.github.io and the framework's About page; verdict that the paper "represents the framework perfectly" – three-tier evidence taxonomy, role taxonomy, multi-AI collaborative architecture, mpmath precision floor, cross-architecture stress-testing, four-layer Stack/Exchanges, and M1–M5 procedural standards all faithfully reflected. Three polish items returned and applied: (i) M1–M5 parenthetical added in §7.2 clarifying that M1–M5 are project-specific refinements of the framework's general principles documented at methodology.html; (ii) Cardano-Ferrari hyphenation normalized across all 25 occurrences; (iii) the verbatim-repeat phrase "the locked illustrative formula as a mathematical object" tightened in §8.3 to "the locked illustrative formula itself" while preserving the anchored statement in the abstract. The full 12-agent pass also re-verified K_{AUD} , $p(L)$, $f_X(1/4)$, Cardano-Ferrari recipe, Galois S_4 classification, and ran the Appendix A script.
- Super Grok (xAI, X chat-side instance, memory-blind to prior framework context) – Stress-testing (independent fresh-eyes pass, 2026-06-01): a memory-isolated read of the updated paper from D.B.'s personal X account, with no link to the framework's prior context or Bee's framing. The pass returned a strong positive verdict on the math and the editorial structure (with specific praise for the EA forced-derivation, the three-constants distinction in CA, the ladder of tractability framing, and the methodology transparency including the §7.3 audit-trail preservation of reader errors). Two minor quibbles surfaced: (i) the Cardano-Ferrari expression's necessary messiness – explicitly noted as "the price of an exact algebraic closure for a generic irreducible quartic" with acknowledgment that the paper does not pretend otherwise (no fix needed; the paper's §4.5–§4.6 already carry this honest framing and Open Thread #1 in §9 explicitly names the

denesting question)); (ii) the suggestion that a formal PARI/GP-style Galois-group computation would tighten the S_4 classification further, marked as "not required for the paper's claims." The PARI/GP suggestion has been recorded as a follow-on tightening at the end of §4.6 (option open for future computer-algebra-community verification). The memory-blind context makes this an especially clean signal: an independent reader judging the paper purely on its own evidence reached the same conclusions the framework-internal reviewers reached.

- Chat-side Opus (Anthropic, fresh-eyes session, 2026-06-01). Independent re-run of Appendix A at dps = 120 reproducing every numerical claim in §3, §4.3, §4.4, §4.5, §4.6, §4.3(ii) to the precision floor with no corrections. Contribution: the §4.3 / §8.2 worked-computation sharpening – direct computation of the residual density $f^*(y_-) = (f_X(y_-) - 0.997738) / (1 - 0.997738) \approx -0.19$ in the locked $m(x)$ of eq. (43) at the critical point y_- ; the $(1 - p) \approx 0.0023$ denominator amplifies the small numerator into a 19% negative residual when read at the dps ≥ 100 resolution this paper uses, a distinction that 1996-era four-decimal-place display conventions did not surface. The sharpening uses only the paper's own definitions (the locked $m(x)$ of §4.1, the f_X of §4.1, the y_- of §4.3) – it is paper-internal mathematics with no source dependency for the verifier. The fresh-Opus reading also noted the implementation block's internal-consistency cross-check (Fishman p. 187 prints both $p = 0.997738$ and $1 - p = 0.002262$ in the implementation setup, matching $2\sqrt{W}$ and $1 - 2\sqrt{W}$ respectively) – strengthening the §4.3 finding that the printed digits live with the implementation side. The new framing in §4.3 / §8.2 presents the discrepancy as a tool-resolution observation rather than a textual flaw. The fresh-eyes position was the source of the catch: a framework-internal reviewer would have accepted " $0.997738 \neq p_{CA}(\text{true})$ " as sufficient and not asked the residual-density question; the fresh-context reader, reading the paper purely on its own terms, asked the sharper question (cf. Methodology Notebook §8.11 on successive review rounds at different time-stages).

Wider readership. Beyond the named contributors above, the framework has been tested continuously across its longer development arc – both in planned stress-test rounds recorded in dedicated research-side files, and in ongoing exploration and challenge passes by fresh AI instances across architectures and devices. Several of the named contributors first joined the work through such passes; many more fresh-instance reads shaped the framework without crossing into named credits in this paper, and several recurring associates appear in the recorded stress-test set rather than in this acknowledgments list. Fresh-eyes and returning-associate impact balance one another in the test landscape – neither is less honest than the other, and both are part of the framework's verification discipline. The framework owes its present shape to every reader-instance – and to the human readers behind them – who engaged honestly with the math along the way.

The role taxonomy distinguishes Conceptualization (asking the right question, seeing the direction) from Discovery (identifying the new structure) from Derivation (proving it formally) from Verification (confirming it independently). Where multiple roles apply, all are listed. The published Methodology document describes this distinction in detail.

References

Primary sources for the algorithms analyzed.

- Ahrens, J. H., & Dieter, U. (1988). Efficient table-free sampling methods for the exponential, Cauchy, and normal distributions. *Communications of the ACM*, 31(11), 1330–1337. DOI: 10.1145/50087.50094. (Original source for Algorithm EA's binary-cell decomposition + rational approximation, and for Algorithm CA's Cauchy rational approximation. The constants quoted in §3.1 and §4.1 are from this paper, as cited by Fishman 1996.)
- Hamilton, K. G. (1998). Algorithm 780: exponential pseudorandom distribution. *ACM Transactions on Mathematical Software*, 24(1), 104–106. DOI: 10.1145/285861.285866. (Independent peer-reviewed implementation of Algorithm EA with corrections to Ahrens and Dieter 1988. The abstract states the work is "based on mathematics that was published by Ahrens and Dieter, but some errors have been corrected." Note: §3's forcing result derives the constants a , b , c from the structural conditions of Algorithm EA rather than importing them from the historical source, so the result holds independently of Hamilton's corrections.)
- Fishman, G. S. (1996). *Monte Carlo: Concepts, Algorithms, and Applications*. Springer Series in Operations Research and Financial Engineering, Springer-Verlag New York. ISBN 0-387-94527-X. (Algorithm EA: pp. 188–189, eqs. 49–52. Algorithm CA: pp. 186–187, eqs. 42–45 illustration + p. 187 implementation. Exact-approximation method: §3.8.)
- Marsaglia, G. (1984). The exact-approximation method for generating random variables in a computer. *Journal of the American Statistical Association*, 79(385), 218–221. DOI: 10.1080/01621459.1984.10477088. (Origin of the exact-approximation framework as a general technique. Both Algorithm EA and Algorithm CA are specific applications of this framework.)

Standard references for Monte Carlo sampling theory.

- Devroye, L. (1986). Non-Uniform Random Variate Generation. Springer-Verlag, New York. (Standard textbook reference for sampling-from-distributions theory. Devroye's treatment of rejection methods and the inversion principle provides broader context for the exact-approximation method as a special case. The author makes the full textbook freely available on his personal page at luc.devroye.org, so this reference is openly verifiable.)
- Knuth, D. E. (1997). The Art of Computer Programming, Volume 2: Seminumerical Algorithms, 3rd edition. Addison-Wesley. (Standard reference for pseudorandom-number-generation theory. Knuth's discussion of generator quality criteria provides context for §3.4–§3.6 cell-length analysis.)
- L'Ecuyer, P. (1988). Efficient and portable combined random number generators. Communications of the ACM, 31(6), 742–751. DOI: 10.1145/62959.62969. (Combined LCGs cited in Fishman §7.13. Not directly used in the present paper; included for context on the broader landscape of pseudorandom-number generators that share Fishman's exact-approximation methodology setting.)

Gap Geometry framework prior results citing K_{AUD} .

- Paper 1: $\sqrt{2} \cdot \ln(2)$ as a Geometric Constant of Binary Information (OSF: 10.17605/OSF.IO/5VZ2R, 2026-01-21). Original derivation of K_{AUD} as a binary-information bound.
- Paper 6: Cross-Domain Signatures of the Boundary Information Invariant (OSF: 10.17605/OSF.IO/RA3UQ, 2026-03-26). K_{AUD} 's appearance across multiple independent domains.
- Paper 7: A Closed Form for the Hodgson-Kerckhoff Tube-Packing Coefficient (OSF: 10.17605/OSF.IO/JBRHQ, 2026-04-03). The HK tube-packing coefficient equals K_{AUD} numerically via the algebraic identity $\operatorname{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$; the derivation route is hyperbolic-trigonometric and is independent of the saturation-theoretic derivation in the present paper.
- Paper 4: The 400/11 Gap Scaling Formula (OSF: 10.17605/OSF.IO/C4GK5, 2026-02-04). Connection between K_{AUD} 's gap $G = 1 - K_{\text{AUD}}$ and the Feigenbaum constant.
- Methodology document: Gap Geometry – Methodology and Verification Standards. <https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/methodology.html>. (Three-tier evidence taxonomy [Theorem]/[Observation]/[Conjecture] used throughout this paper.)

Computational tools.

- Johansson, F., et al. (2013–present). mpmath: a Python library for arbitrary-precision floating-point arithmetic. <http://mpmath.org>. (Used at $\text{dps} \geq 100$ –1000 throughout this paper. Specific calls: findroot, polyroots, pslq, identify, plus standard arithmetic at high precision.)
- Meurer, A., et al. (2017). SymPy: symbolic computing in Python. PeerJ Computer Science, 3, e103. (Used by Grok in §4.5 Q3 R2 for the independent quartic-solver verification.)

Appendix A – Verification script

The following self-contained mpmath script reproduces all numerical claims in §3, §4, and §5 at $\text{dps} = 100$. Reading the printed differences as ≈ 0 (or $\leq 10^{-98}$ in magnitude) is the expected output.

```
"""
Verification script for "Saturation Constants of the Exact-Approximation Method"
Reproduces all numerical claims at mpmath dps = 100.
Expected: every Difference line is ~0 or  $\leq 10^{*(-98)}$ .
"""
from mpmath import mp, mpf, pi, sqrt, sinh, ln, findroot, polyroots
mp.dps = 100

# === §3 verification:  $K_{\text{AUD}} = \sqrt{2} \cdot \ln(2)$  for Algorithm EA ===

K_AUD = sqrt(2) * ln(2)
p_at_ln2_via_sinh = ln(2) / (2 * sinh(ln(2) / 2))
print("=== §3 (Algorithm EA,  $K_{\text{AUD}}$  identity) ===")
print(f"   $K_{\text{AUD}} = \sqrt{2} \cdot \ln(2)$       = {K_AUD}")
print(f"   $p(\ln 2)$  via (3.4)      = {p_at_ln2_via_sinh}")
print(f"  Difference                    = {p_at_ln2_via_sinh - K_AUD}")
b_plus = 2 + sqrt(2)
```

```

print(f"  b2 - 4b + 2 at b = 2 + √2: {b_plus**2 - 4*b_plus + 2}")

# === §4 setup: locked Cauchy m(x) ===

s = pi**2 # = π2
a_coef = 2 / pi
b_coef = pi - 8 / pi

def m_y(y):
    return y * (a_coef / (mpf('0.25') - y**2) + b_coef)

def mprime_y(y):
    return a_coef * (mpf('0.25') + y**2) / (mpf('0.25') - y**2)**2 + b_coef

def fX(y):
    return (1 / pi) * mprime_y(y) / (1 + m_y(y)**2)

# === §4.4 verification: closed form for f_X(1/4) ===

cc_form = 16 * (9 * pi**2 + 88) / (9 * pi**4 + 192 * pi**2 + 64)
fx_quarter = fX(mpf('0.25'))
print("\n=== §4.4 (closed form for f_X(1/4)) ===")
print(f"  16(9π2 + 88) / (9π4 + 192π2 + 64) = {cc_form}")
print(f"  Direct f_X(1/4) = {fx_quarter}")
print(f"  Difference = {fx_quarter - cc_form}")

# === §4.5 verification: quartic and Cardano-Ferrari closed form ===

# Quartic coefficients per §4.5, eq. (4.6)
c4 = (s - 8)**3
c3 = -4 * (s - 8)**2 * (s - 12)
c2 = 6 * s**3 - 120 * s**2 + 608 * s
c1 = -4 * s**3 + 32 * s**2 + 64 * s
c0 = s**3 - 96 * s

# Real roots of P(t) = 0 in (0, 1)
roots = polyroots([c4, c3, c2, c1, c0])
# Filter to real roots and pick the small one in (0, 0.5) – this is t_*
real_roots = sorted([r.real for r in roots if abs(r.imag) < mpf('1e-50')])
t_star = [r for r in real_roots if 0 < r < mpf('0.5')][0]
y_star_direct = sqrt(t_star) / 2
p_CA_true = fX(y_star_direct)

print("\n=== §4.3 / §4.5 (true minimum via direct quartic-root) ===")
print(f"  t_* = {t_star}")
print(f"  y_* = √t_*/2 = {y_star_direct}")
print(f"  p_CA(true) = f_X(y_*) = {p_CA_true}")

# Verify Cardano-Ferrari recipe (eqs. 4.7–4.11 in §4.5)
A3 = c3 / c4
A2 = c2 / c4
A1 = c1 / c4
A0 = c0 / c4
p_d = A2 - 3 * A3**2 / 8
q_d = A3**3 / 8 - A3 * A2 / 2 + A1
r_d = -3 * A3**4 / 256 + A2 * A3**2 / 16 - A3 * A1 / 4 + A0

# Resolvent cubic via Cardano (real root) – two computations for verification:
# (a) findroot reference (used by Ferrari step below)
y_F = findroot(
    lambda y: y**3 + 2 * p_d * y**2 + (p_d**2 - 4 * r_d) * y - q_d**2,
    mpf('10'),
)

# (b) explicit Cardano formula (4.9)-(4.10) – tests the corrected (4.10)
from mpmath import cbrt
P = -p_d**2 / 3 - 4 * r_d # (4.10) corrected
Q = -2 * p_d**3 / 27 + 8 * p_d * r_d / 3 - q_d**2 # (4.10) corrected
disc = Q**2 / 4 + P**3 / 27
y_F_cardano = cbrt(-Q/2 + sqrt(disc)) + cbrt(-Q/2 - sqrt(disc)) - 2*p_d/3
print("\n=== §4.5 (4.9)-(4.10) explicit Cardano formula verification ===")

```

```

print(f" y_F via findroot (reference): {y_F}")
print(f" y_F via (4.9)-(4.10) Cardano: {y_F_cardano}")
print(f" Difference : {y_F_cardano - y_F}")

# Ferrari step
C_b = (y_F + p_d) / 2 + q_d / (2 * sqrt(y_F))
D_b = y_F - 4 * C_b
u_star = (sqrt(y_F) - sqrt(D_b)) / 2
t_star_CF = u_star - A3 / 4
y_star_CF = sqrt(t_star_CF) / 2
p_CA_CF = fX(y_star_CF)

print("\n=== $4.5 (Cardano-Ferrari closed form, eqs. 4.7-4.11) ===")
print(f" y_F (resolvent root) = {y_F}")
print(f" D_b = {D_b}")
print(f" u_* = {u_star}")
print(f" t_* via Cardano-Ferrari = {t_star_CF}")
print(f" y_* via Cardano-Ferrari = {y_star_CF}")
print(f" p_CA(true) via CF route = {p_CA_CF}")
print(f" Difference t_* (CF - direct) = {t_star_CF - t_star}")
print(f" Difference y_* (CF - direct) = {y_star_CF - y_star_direct}")
print(f" Difference p_CA (CF - direct) = {p_CA_CF - p_CA_true}")

# === $4.3 (ii): verify  $2\sqrt{W} = 0.997738$  in Algorithm CA's implementation ===

W = mpf('0.2488702280083841') # Implementation constant from Fishman p. 187
fast_path_prob = 2 * sqrt(W)
print("\n=== $4.3 (ii) (implementation correctness on p. 187) ===")
print(f"  $2\sqrt{W}$  with  $W = 0.2488702280083841$ : {fast_path_prob}")
print(f" Fishman printed 0.997738 differs by: {fast_path_prob - mpf('0.997738')}")

```

Expected output: every "Difference" line shows a value $\leq 10^{-98}$ in magnitude (essentially zero at mp.dps = 100), confirming all closed-form claims to displayed precision.

Document Links

The framework

- Main OSF project (immutable timestamps, version history, all papers):
osf.io/zx4g7 · DOI: 10.17605/OSF.IO/ZX4G7
- Framework overview (reading order, methodology, all papers):
gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html
- Direct downloads (current PDF and text files for every paper):
gap-geometry.github.io/sqrt2-ln2-geometric-constants-/direct-documents.html

This paper

- DOI: 10.17605/OSF.IO/6QZRB
- OSF project: osf.io/6qzrb

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